Chapter 5
Muscles

Abstract Overviews of the types of muscles in the body and their macroscopic structure are first presented, followed by models of muscle performance from macroscopic, microscopic, and nanoscopic perspectives. This includes analysis of forces generated by muscles as a function of their length and contraction speed, and how they depend on passive muscle materials and tendons. The discussion of the sliding filament model and sarcomeres includes the role of the elastic material titin. Muscle coordination, training, development, and fatigue are also addressed.

In this chapter we will discuss the motors of the body, which are the muscles. There are three types of muscles in the body (Fig. 5.1): (1) *Skeletal muscles* are involved in skeletal motion, lip motion, and in eyelid and eyeball motion. They consist of long cylindrical cells that appear striated, have many nuclei, and are under conscious control (*voluntary muscles*). Although skeletal muscles can be consciously controlled, some move without conscious effort, as in the diaphragm. (2) *Cardiac muscle* comprises most of the heart. These muscle cells are striated, have one centrally located nucleus, and often branch; when two cardiac cells meet they form an intercalated disc. (3) *Smooth muscles* line the walls of blood vessels and the digestive and urogenital organs, to help advance and control flow (see Figs. 7.22 and 8.14). They are elongated cells with tapered ends and no striation, have a single, centrally located nucleus, and are not under conscious control (*involuntary muscles*), as is also true for cardiac muscle. We will concentrate on skeletal muscles; many of the properties of cardiac and smooth muscles are similar. Some differences between cardiac and skeletal muscles are described in Chap. 8.

We will learn how muscles work and develop macroscopic, microscopic, and nanoscopic models of them [50, 52, 53, 60, 70, 82]. We will also re-examine some of the assumptions we made in earlier chapters involving muscle forces. In the next chapter we will learn more about how energy is used in muscles.

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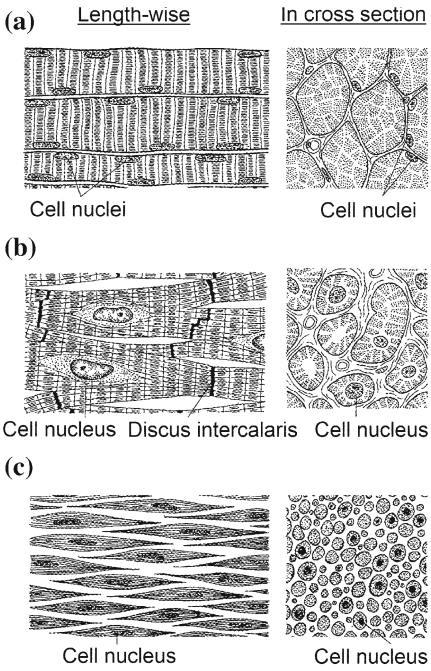


Fig. 5.1 **a** Skeletal, **b** cardiac, and **c** smooth muscle cells. (From [67])

5.1 Skeletal Muscles in the Body

Figure 1.8 shows many of the main skeletal muscles in the body.

Skeletal muscles account for ∼43% of the typical body mass, or ∼30 kg of a 70 kg person. At rest, they use ∼18% of the body energy consumption rate (which is called the basal metabolic rate or BMR); they use much more during activities, such as motion. Of the total energy "burned" by skeletal muscle, only ∼25% is used for work, and this is the muscular efficiency. The other ∼75% is released as heat. (This inefficiency of the body does have a positive purpose—it is a consistent and very important source of body heat.)

The maximum muscle force F_M or tension T that a muscle can develop is k_MPCA, where PCA is the physiological cross-sectional area of the muscle. The range of the maximum values of k_M is 20–100 N/cm^2 during isometric (constant length) conditions; the larger values are for muscles with the pinnate fiber structure that is described below. (See Fig. 5.46 below and Problem 5.2.) In the quadriceps, the forces are up to ∼70 N/cm^2 during running and jumping and ∼100 N/cm^2 under isometric conditions. We will discuss this more later.

Countermovements before sudden motion are known to increase athletic performance [63]. Quickly squatting before squat jumps leads to higher jumps. It is not known whether this pre-stretching allows more time for muscles to become active, optimizes sarcomere lengths and forces (see below), induces the activation of

Table 5.1 Percent PCA of muscles crossing the hip joint

Muscle	%PCA
Iliopsoas	9
Sartorius	1
Pectineus	1
Rectus femoris	7
Gluteus maximus	16
Gluteus medius	12
Gluteus minimus	6
Adductor magnus	11
Adductor longus	3
Adductor brevis	3
Tensor fasciae latae	1
Biceps femoris (long)	6
Semitendinosus	3
Semimembranosus	8
Piriformis	2
Lateral rotators	13

From [81], data from [76]

other muscles, or stores elastic energy in tendons, ligaments or the elastic components of muscles. This countermovement may be the reason why balls can be thrown faster with a windup during baseball pitching, and this might be due to the storage and subsequent release of elastic energy.

Sometimes the estimate that a maximum of ~250–500 W are generated per kg of active muscle for isotonic conditions (constant tension) is used in calculations of motion (and 1.75× this with pre-stretching) [13, 25, 78, 79]. Because this is averaged over the areas, lengths and speeds of the muscles that cause motion, we will usually use the muscle force per unit area in analysis.

Many muscles that cross the hip, knee, and ankle joints are important in locomotion. The relative significance of these leg muscles depends on the locations of their points of origin and insertion, lengths, and their PCAs. Quite a few of these muscles are depicted in Figs. 3.3, 3.4, 3.5. (Analogous sets of muscles are involved in controlling our arms and hands, as seen below in Figs. 5.22 and 5.23.) The relative PCAs are given in Tables 5.1, 5.2, 5.3. In general, longer muscles enable larger angles of rotation about joints. The larger the muscle PCA, the more the muscle strength. Muscles with smaller PCAs can be important in providing stability.

As is clear from Figs. 1.8 and 3.2 and from Tables 5.1, 5.2, 5.3, many of these leg muscles pass over more than one joint. Figure 5.2 shows the three such major biarticulate (two joint) muscles in the leg. The forces exerted by the muscles on the bones at the points of origin and insertion are the same (Newton's Third Law), but the torques are different at the proximal and distal joints because the moment

Table 5.2 Percent PCA of muscles crossing the knee joint

Muscle	%PCA
Gastrocnemius	19
Biceps femoris (small)	3
Biceps femoris (long)	7
Semitendinosus	3
Semimembranosus	10
Vastus lateralis	20
Vastus medialis	15
Vastus intermedius	13
Rectus femoris	8
Sartorius	1
Gracilis	1

From [81], data from [76]

Table 5.3 Percent PCA of muscles crossing the ankle joint

Muscle	%PCA
Soleus	41
Gastrocnemius	22
Flexor hallucis longus	6
Flexor digitorum longus	3
Tibialis posterior	10
Peroneus brevis	9
Tiabialis anterior	5
Extensor digitorum longus	3
Extensor hallucis longus	1

From [81], data from [76]

arms are different. As seen in Fig. 5.2, the gastrocnemius is a knee flexor and ankle plantarflexor (extensor), with the torque about the latter joint greater because of its larger moment arm (5 cm vs. 3.5 cm). During stance the net effect of this muscle is to cause the leg to rotate posteriorly and to prevent the knee from collapsing. The hamstrings are extensors of the hip and flexors of the knee, with hip extension having twice the moment of knee flexion. During stance, this causes the thigh to rotate posteriorly and prevents the knee from collapsing. The rectus femoris of the quadriceps is a hip flexor and knee extensor, with a slightly larger moment about the hip. However, the major action of the quadriceps is knee extension because 84% of the quadriceps PCA is from other muscles, the uniarticulate knee extensors (and all of these muscles fire simultaneously). The net effect of these three major biarticulate muscles is the extension of all three leg joints, and they help prevent against collapse due to gravity. Coordinated motion of these muscles, such as the lengthening of one of these muscles and shortening of another, is needed to achieve the full range of

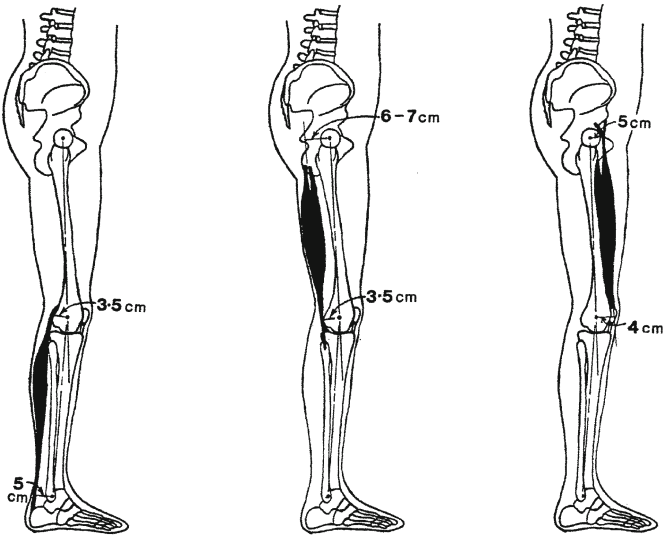


Fig. 5.2 Three major biarticulate muscles of the leg, from left to right the gastrocnemius, hamstrings, and rectus femoris, along with moment-arm lengths about the joints at their proximal and distal ends. (From [81]. Reprinted with permission of Wiley)

rotation of these joints because of the limited change of muscle length of any one of them [81].

5.1.1 Types of Muscle Activity

Muscles can be activated under a wide range of conditions. When the muscle length changes, the angle of the joint for this inserted muscle changes, and there is motion at the end of the bone emanating from the joint. Because there is motion, mechanical work is done. This can be called "dynamic" work. When the muscle length does not change, i.e., *isometric* conditions, there is no rotation of the joint and no mechanical work is done. This is important because tension is still supplied by the muscle to resist outside forces, as needed in holding objects or standing upright. Energy is still expended by the muscle (for such "static" work) to produce the tension that resists the load.

Mechanical work is performed with joint motion for nonisometric contractions. In *concentric* contractions, the muscle develops enough tension to overcome the load and the muscle length shortens, causing joint movement. In ascending stairs the quadriceps contract and the leg straightens as a result of this concentric contraction. In *eccentric* contractions, the muscle does not develop enough tension to overcome the load and the muscle length still lengthens, sometimes slowing joint movement (on purpose). In descending stairs the quadriceps are activated but they still

extend during this essential controlled (eccentric) braking of knee flexion in fighting gravity. Therefore, the same flexor muscles that contract concentrically during flexion can contract eccentrically during extension to decelerate the extension. Concentric contractions are said to do *positive (mechanical) work*, while eccentric contractions do *negative work*.

During *isokinetic* contraction, the velocity of muscle shortening or lengthening, and consequently also the angular speed of the joint, are constant. During *isoinertial* contraction, the resistive load on the muscle—due to the gravity force, applied forces, etc.—is constant. During *isotonic* contraction, the tension is constant. (This is an idealized condition because the muscle tension actually changes with length.)

In elbow flexion the biceps brachii and brachialis muscles contract and they are the *agonists* or prime movers of the action. The brachioradialis is a *synergist* muscle in the motion; such synergist muscles assist the motion and sometimes add fine tuning. The triceps brachii are the *antagonists* of this action, and oppose the prime movers.

During walking and running the quadriceps act eccentrically during early stance to prevent the collapse of the knee angle, and then concentrically to extend the knee as the leg rotates over the foot in midstance (and this is more so in walking). The ankle plantarflexors fire eccentrically during stance to help advance the leg rotation over the foot controllably, and then concentrically to assist push off.

5.2 The Structure of Muscles

Muscles have a fiber structure with successive levels of fiber-like substructures called fasciculi (fuh-sik'-you-lie). (A single one is a fasciculus (fuh-sik'-you-lus).) On a macroscopic basis, these fibers are arranged in one of several ways. This is illustrated in Fig. 5.3. It appears that the forces generated by individual muscle fibers and muscles can be added independently [65, 66].

In *fusiform* muscles the muscle fibers are parallel and they narrow and blend into tendons that attach to the skeleton. In *parallel* muscles the fibers are also parallel. They tend to be long, such as the sartorius—the longest muscle in the body, which spans from the hip to the tibia (Figs. 1.8 and 5.4). (The fibers are also parallel in the

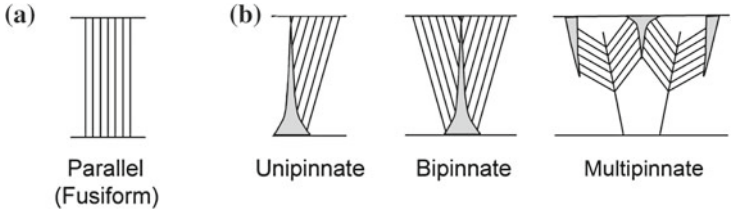


Fig. 5.3 Types of fiber arrangements in skeletal muscles, with **a** parallel and **b** pinnate fibers. (Based on [53])

rectus abdominis (the "abs").) We will usually not differentiate between fusiform and parallel muscles because of their similar arrangement of muscle fibers. These muscles consist of many sarcomere components in series (see below), and as such they can become much shorter to produce much movement of bones for motion about joints, and they can do so quickly. In *unipinnate* muscles, parallel muscle fibers attach to tendons at an angle, such as with the flexor policis longus (poe-lee'-cis) in the lower hand/thumb and the extensor digitorium longus (di-gi-tor'-ee-um) in the lower leg/foot. In *bipinnate* muscles, such parallel muscle fibers attach to a tendon in two different directions, such as with the rectus femoris (fe-more'-is) in the thigh. These *pinnate* (or pennate or pinnation) structures resemble the structure of feathers and of some leaves on branches. The fibers attach on the central tendon at several angles in the *multipinnate* deltoid (shoulder) muscles. Many short muscle fibers can attach to such short tendons in pinnate muscles, leading to larger forces than for parallel muscles and more movement, but with less efficient use of the muscles forces (see below and Problem 5.17). The orbicularis oculi (or-bee-queue-lar'-is ok'-you-lie) about the eye are *circular* muscles, in which the muscle fibers are in a circle about the object. The pectoralis major (pec-tor-al'-is) in the upper chest (the "pects") is a *convergent* muscle; such muscles are broad at their origin and narrow at insertion, leading to large forces near the insertion point. Examples of each of these muscle types in the body are shown in Fig. 5.4.

The way the forces from muscle fibers add to give the total force on the attaching tendon differs for these different muscle structures. In parallel, fusiform muscles, all the force of the fibers is transmitted to the tendon. In pinnate muscles, the fibers are attached to the tendon at an angle θ, and only $F\cos\theta$ of the force $\mathbf{F}$ of each fiber is effectively transmitted. While this is a distinct disadvantage of the pinnate design, it has other relative advantages. Because the geometry allows fibers to attach along part of the length of the tendon (Fig. 5.5), many more fibers can be attached to the tendon. Also, this geometry allows the central tendon to move a longer distance than in the fusiform scheme, so the bones attached to the tendon can move more. This overcomes the limited change in length in the muscles that can limit the range of angular motion of bones about joints. These and related issues are addressed in Problem 5.17. The mass, fiber length, PCA, and pinnation angle θ for several muscles are given in Table 5.4.

The microscopic components of the fibers are illustrated in Figs. 5.6 and 5.7, where the muscle belly of a fusiform muscle is seen to be composed of many parallel *fasciculi*. Each fasiculus is composed of many parallel *muscle fibers* (or *muscle cells*). Each muscle fiber is composed of many parallel *myofibrils*. Each myofibril is composed of *myofilaments* arranged into $\sim$2–3 μm long units called *sarcomeres* (sar'-koe-meres). There are thick myofilaments with a serial arrangement of many *myosin* (my'-oh-sin) molecules and thin myofilaments with globules of *F-actin* (or *actin*) molecules that form twisting strands, which are surrounded by two other proteins: *tropomyosin*—which forms strands that twist about the actin strands—and *troponin-T*—which attaches at regular intervals to the actin and tropomyosin strands. The interaction between the myosin and actin proteins on adjacent myofilaments is the

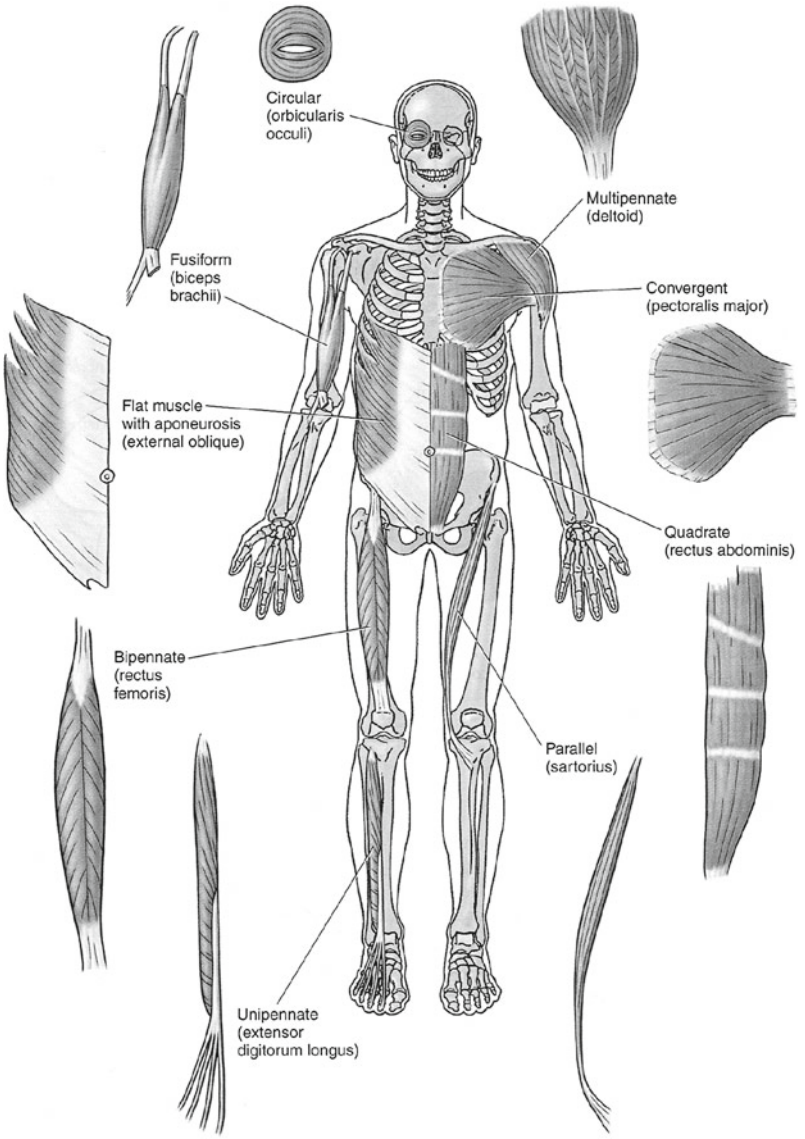


Fig. 5.4 Examples of muscle fiber arrangements in the body. (From [58]. Used with permission)

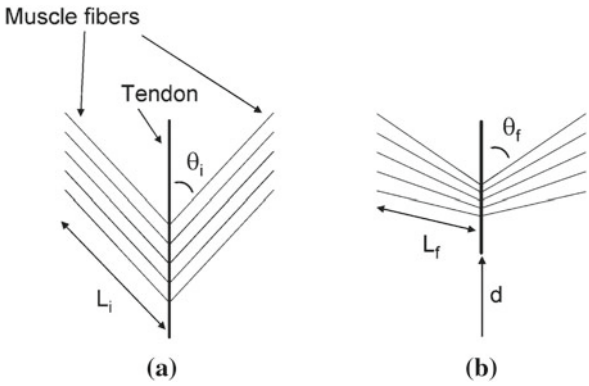


Fig. 5.5 Muscle fibers in pinnate muscles **a** before and **b** after fiber contraction with accompanying tendon movement

Table 5.4 Properties of some muscles

Muscle	Mass (g)	Fiber length (cm)	PCA (cm^2)	Pinnation angle (°)
Sartorius	75	38	1.9	0
Biceps femoris (long)	150	9	15.8	0
Semitendinosus	75	16	4.4	0
Soleus	215	3.0	58	30
Gastrocnemius	158	4.8	30	15
Tibialis posterior	55	2.4	21	15
Tibialis anterior	70	7.3	9.1	5
Rectus femoris	90	6.8	12.5	5
Vastus lateralis	210	6.7	30	5
Vastus medialis	200	7.2	26	5
Vastus intermedius	180	6.8	25	5

From [81], data from [76]

fundamental, chemical-induced, force-producing interaction in the muscle, which involves the hydrolysis of ATP and the change in conformation of the myosin.

In the large limb of an adult the muscle fibers are ~50 μm in diameter. This diameter can double with weight training. Depending on the type of muscle, there are $\sim 10^4$–10^7 muscle fibers per muscle (Table 5.5). There are $\sim 1 \times 10^4$–1.7×10^5 sarcomeres per muscle fiber, depending on the type of muscle (Table 5.6).

The electron micrograph in Fig. 5.7 shows the banded myofilament structure that is also sketched in the figure. The A band is a wide dark, anisotropic region, while the I band is a wide light, isotropic region. There is a lighter H zone or band in the middle of the dark A band, and a darker M line in the middle of the H zone. The dark Z lines run through the light I bands.

Each sarcomere is bound between the adjacent Z lines. The dark A band in the center of each sarcomere consists of thick (myosin) myofilaments (or thick filaments), which are connected to each other in the central M line within the H zone. They are

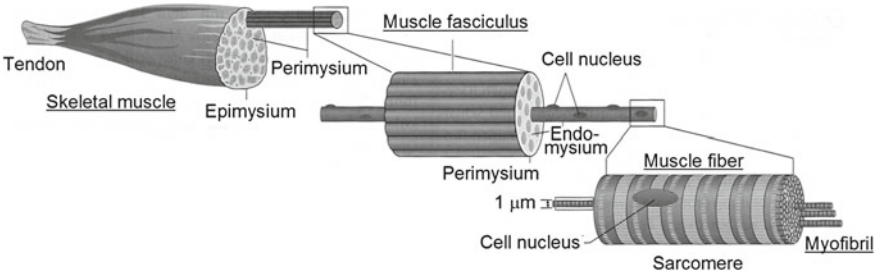


Fig. 5.6 Organization of skeletal muscles, down to the myofibril level. See Fig. 5.7 for the structure of the myofibril. (From [68])

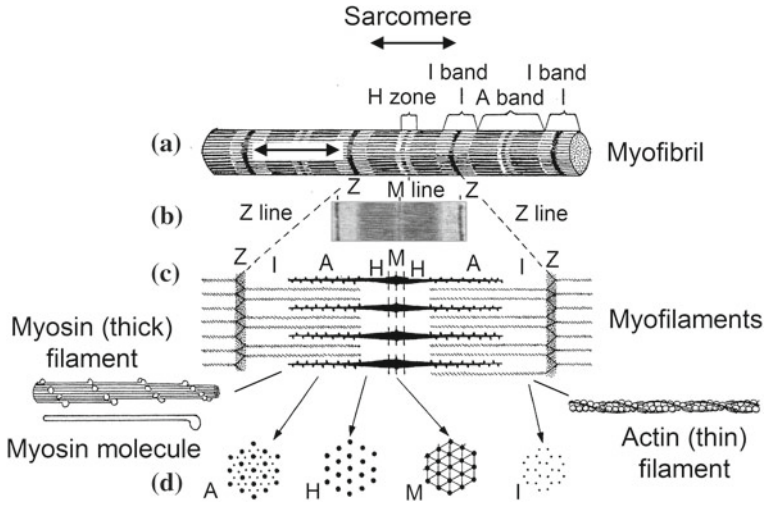


Fig. 5.7 **a** Structure and substructure of the myofibril shown in Fig. 5.6, with the banded structure of sarcomeres (whose periodicity is denoted by the arrows). **b** The scanning electron micrograph of skeletal muscle shows the structure of bands and lines in sarcomeres. This is associated with the thick and thin filaments in (**c**) from the longitudinal perspective of (**a**) and (**b**), the transverse cross-section is seen in (**d**). (From [67])

overlapped to some degree by the thin (actin) myofilaments (or thin filaments). There are thick filaments but no thin filaments in the central H zone. The light I band regions next tox the Z lines are the thin myofilaments in regions where they do not overlap the thick myofilaments; they are bound to each other at the Z line. This is shown schematically in Fig. 5.7.

Table 5.5 Number of muscle fibers in human muscles

Muscle	Number of muscle fibers
First lumbrical	10,250
External rectus	27,000
Platysma	27,000
First dorsal interosseous	40,500
Sartorius	128,150
Brachioradialis	129,200
Tiabialis anterior	271,350
Medial gastrocnemius	1,033,000

Using data from [52]

Table 5.6 Number of sarcomeres in human muscles

Muscle	Number of sarcomeres per fiber ($\times 10^4$)		
	Person I	Person II	Person III
Tibialis posterior	1.1	1.5	0.8
Soleus	1.4	–	–
Medial gastrocnemius	1.6	1.5	1.5
Semitendinosus	5.8	6.6	–
Gracilis	8.1	9.3	8.4
Sartorius	15.3	17.4	13.5

Using data from [52]

5.3 Activating Muscles: Macroscopic View

Electrical stimuli lead to twitches in the muscles that temporarily increase the force exerted by them. These twitches are delayed by about 15 ms after the electrical stimulus. They peak ~40 ms later, and then decay to zero ~50 ms later (Fig. 5.8a, Table 5.7). The shape of a twitch can be modeled as

$$F(t) = F_0 \frac{t}{T} \exp(-t/T) \tag{5.1}$$

with twitch time T.

By increasing the frequency of these stimuli and consequently of the twitches, there is an increase in the force exerted by the muscle (Fig. 5.8b, c). At a large enough frequency the twitches overlap to produce an almost steady level of force called *unfused tetanus,* and at an even higher frequency they produce a force that is constant in time called *tetanus* (Fig. 5.9). Roughly 50–60 electrical shocks per second are required to fully tetanize mammalian muscles at room temperature. This varies from about ~30/s for the soleus muscle to ~300/s for eye muscles. (Also see Fig. 5.11 below.)

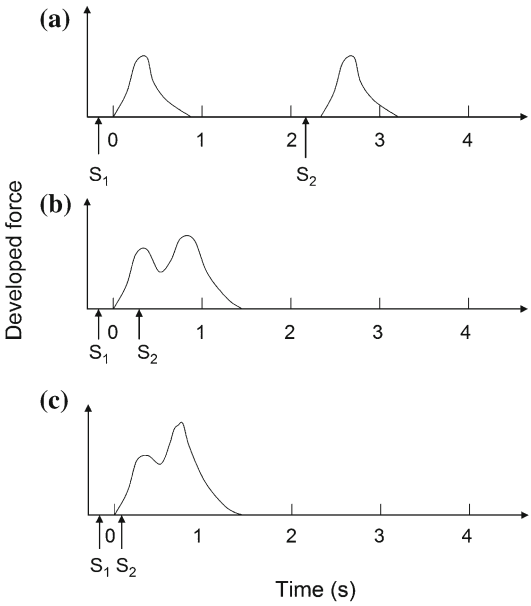


Fig. 5.8 Two twitches stimulated at S_1 and S_2, with various delays. (Based on [53])

Table 5.7 Twitch time T, in ms

Muscle	Typical mean time	Range of times
Triceps brachii	44.5	16–68
Biceps brachii	52.0	16–85
Tibialis anterior	58.0	38–80
Soleus	74.0	52–100
Medial gastrocnemius	79.0	40–110

Using data from [81]

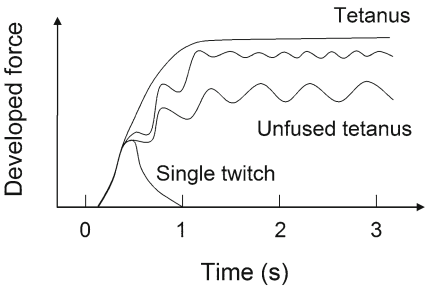


Fig. 5.9 A twitch, and then a series of twitches leading to unfused tetanus, and at an even higher frequency leading to tetanus. (Based on [53])

There are three different types of muscle fibers, which differ in how fast they contract and their resistance to fatigue. They appear in different types of muscles in varying proportions (Fig. 5.10). For a given muscle type, their relative concentrations can be different in different people. Slow-twitch (ST) red fibers (Type I) have a long

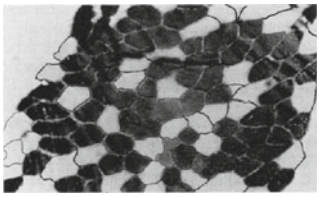


Fig. 5.10 Photomicrograph of stained muscle cells from a human vastus lateralis muscle, showing ST Type I (*dark*) and FT Type II (*lightly stained*) cells, with *fine lines* showing boundaries added. (From [20]. Used with permission)

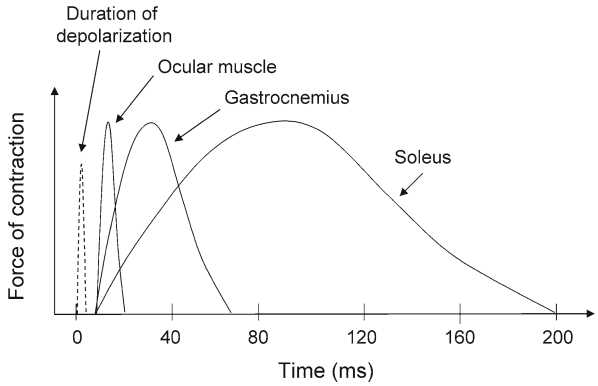


Fig. 5.11 Duration of isometric responses for different muscles with different FT and ST muscle fibers. (Based on [26])

contraction time (∼110 ms) and are very resistant to fatigue because they are aerobic, i.e., they use oxygen to produce ATP. These fibers are red because they have blood to supply oxygen. Fast-twitch (FT) fibers generally can create more force than ST fibers, and reach a peak tension in less time (∼50 ms). FT red intermediate fibers (Type IIA) have a relatively short contraction time, an intermediate fatiguing rate, and aerobic generation of ATP. FT white fibers (Type IIB) have a short contraction time and fatigue quickly because they use anaerobic (i.e., no oxygen) processes to produce ATP. One of the reasons for the faster response of the fast twitch muscle is the larger neuron exciting it.

The "average" muscle has roughly 50% ST fibers and 25% red and white FT fibers. The contraction time of a muscle depends on the proportion of FT and ST fibers. Figure 5.11 shows the twitch response for three types of muscles. The ocular muscles in the eye have mostly FT muscles and a response time of ∼1/40 s. The gastrocnemius (gas-trok-nee'-mee-us) muscle has many FT muscles and a response time of ∼1/15 s. The soleus (soh'-lee-us) muscle has many ST muscles and a response time of ∼1/5 s.

All of this makes sense in terms of the body's needs. The eye muscles must be fast for our needed rapid eye reflex. The gastrocnemius and soleus muscles in the

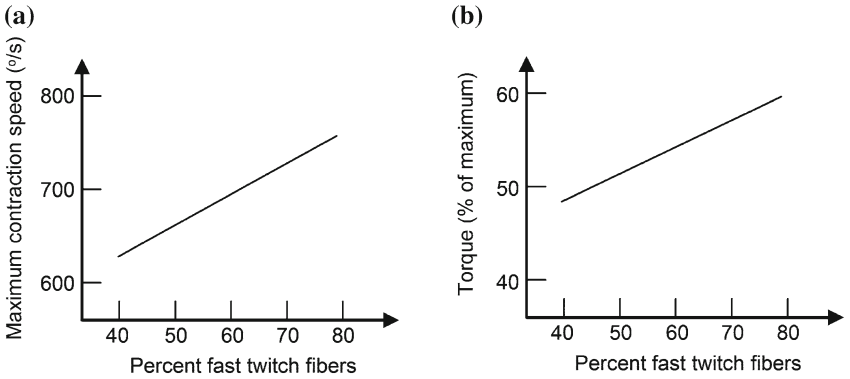


Fig. 5.12 Knee extension performance versus percentage of FT fibers in the knee. (Based on [21] and [73])

upper calf are both connected to the Achilles (calcaneal) tendon in the foot, which is connected to the calcaneus bone in the heel. The soleus muscle is a broad calf muscle that is deep relative to the medial and lateral heads of the gastrocnemius muscle (see Figs. 1.8b and 3.5). The soleus muscle is used more for standing and stability, for which endurance and resistance to fatigue are important and a fast response is not very important. The gastrocnemius muscle is used mostly for jumping and running, for which a fast response is necessary, even at the expense of endurance. The relative fraction of ST and FT muscles in the knee for trained athletes is depicted in Fig. 5.12 and Table 5.8. Clearly, athletes in sports that require endurance—such as marathon runners—have a higher fraction of ST muscle fibers in muscles used in running, while those in sports that require speed and strength—such as sprint runners and weightlifters—have a higher fraction of FT muscles in those muscles heavily used in these activities. There is a similar correlation for the cross-sectional areas of these different muscle fibers.

5.3.1 Mechanical Model of the Active State of Muscles

We now develop a macroscopic model of muscles in the active state due to electrical stimulation by using the type of spring–dashpot models we employed to characterize viscoelasticity in Chap. 4. This model includes springs and dashpots to account for the passive properties of the muscle and tension generators to characterize the active state. While it mathematically characterizes the mechanical features of skeletal muscles acting against loads quite well, the individual components of the model may or may not describe muscle properties microscopically. For example, the viscosity represented by the dashpot does not model the effect of the viscosity of the fluid in a muscle very well. The parallel elastic element may, in fact, be due to the

Table 5.8 Percentages of ST and FT muscle fibers in selected muscles in male (M) and female (F) athletes, along with the cross-sectional areas of these muscle fibers

Athlete	Gender	Muscle	%ST	%FT	ST area (μm^2)	FT area (μm^2)
Sprint runners	M	Gastrocnemius	24	76	5,878	6,034
	F	Gastrocnemius	27	73	3,752	3,930
Distance runners	M	Gastrocnemius	79	21	8,342	6,485
	F	Gastrocnemius	69	31	4,441	4,128
Cyclists	M	Vastus lateralis	57	43	6,333	6,116
	F	Vastus lateralis	51	49	5,487	5,216
Swimmers	M	Posterior deltoid	67	33	–	–
Weightlifters	M	Gastrocnemius	44	56	5,060	8,910
	M	Deltoid	53	47	5,010	8,450
Triathletes	M	Posterior deltoid	60	40	–	–
	M	Vastus lateralis	63	37	–	–
	M	Gastrocnemius	59	41	–	–
Canoeists	M	Posterior deltoid	71	29	4,920	7,040
Shot-putters	M	Gastrocnemius	38	62	6,367	6,441
Nonathletes	M	Vastus lateralis	47	53	4,722	4,709
	F	Gastrocnemius	52	48	3,501	3,141

From [80]

sarcolemma, which is the outer membrane surrounding the muscle fiber. The highly elastic protein, titin (see Fig. 5.18 below), forms a net-like structure about the thick and thin filaments, and may also contribute to this element. The hinge regions of myosin may contribute to a series elastic element. (In cardiac muscle, it is believed that connective tissue is the major part of the parallel elastic element.)

Figure 5.13a shows a mechanical model with a unit composed of a tension generator $T_0(x, t)$ in parallel with a dashpot with viscosity c and a spring with spring constant k_{parallel}, that is in series with another spring with spring constant k_{series}. The total length of the muscle is $x^T = x^E + x$, where x^E is the equilibrium length and x is the displacement.

One can show that this model (which is also depicted in Fig. 5.14a) is mathematically equivalent to the model shown in Fig. 5.14b in which there is a unit with a spring in series with a tension generator in parallel with a dashpot, and this unit is in parallel with another spring. They are mathematically equivalent in the sense that combinations of the components give the same model predictions. The spring constants and viscosity coefficients in the two models are not equal to each other, but are related to each other (see Problem 5.10). This representation of the model in (a) is exactly the same as the Kelvin/standard linear model, except that the dashpot is in parallel with the tension generator.

We will examine the state of the muscle by solving a slightly simpler model, with only the spring with k_{series} (now called k for simplicity) in Fig. 5.15. Without

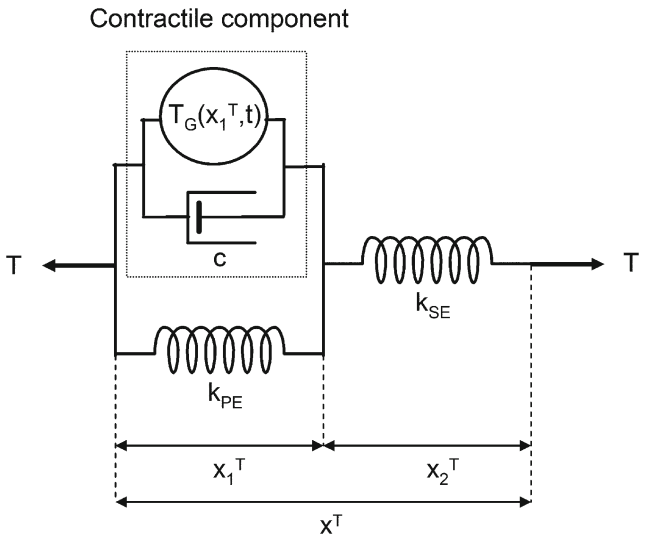


Fig. 5.13 Complete active state muscle model. (Based on [53])

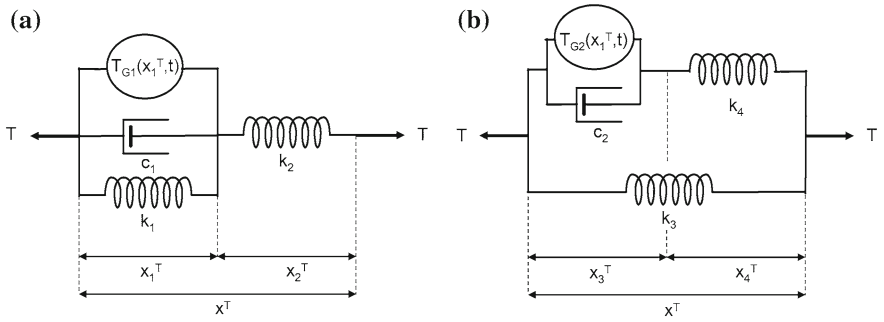


Fig. 5.14 Equivalent active state muscle models. The model in part **a** is the same as that in Fig. 5.13. (Based on [53])

the tension generator, this reduces to the Maxwell model. The tension generator supplies a tension that could depend on muscle length x and time t, $T_G = T_G(x, t)$. We will also assume that the total length of the muscle does not change, i.e., isometric conditions. This means that $x^T = x_1^T + x_2^T$ is constant, where x_1^T is the length of the dashpot/tension generator and x_2^T is the length of the spring. For isometric conditions, we know that $x(t) = 0$. Consequently, the tension generator $T_G = T_G(t)$. We will say this generator supplies the tension T_0 for specific durations of time.

The total length of the dashpot/tension generator can be subdivided into the equilibrium length and the displacement $x_1^T = x_1^E + x_1$; for the spring we see that $x_2^T = x_2^E + x_2$. The force generated across the muscle is $T = T(t)$. This is equal to the tension across the spring

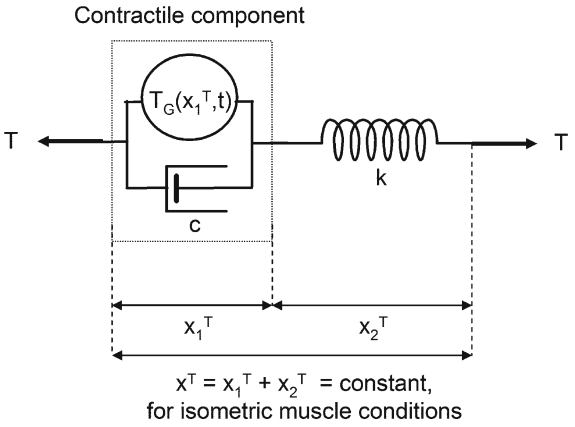


Fig. 5.15 Simpler active state muscle model. (Based on [53])

$$T = kx_2 \tag{5.2}$$

and is also equal to the sum of the tensions across the tension generator and dashpot, and so

$$T = T_{\mathrm{G}} + c\,\frac{\mathrm{d}x_1}{\mathrm{d}t}. \tag{5.3}$$

(Refer to the discussions of the Maxwell and Voigt models of viscoelasticity in Chap. 4 for a more detailed explanation.)

Because $x^{\mathrm{T}} = x_1^{\mathrm{T}} + x_2^{\mathrm{T}}$ is constant, we see that $\mathrm{d}x^{\mathrm{T}}/\mathrm{d}t = 0$. Also, because the equilibrium distances are constant

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\mathrm{d}x_1}{\mathrm{d}t} + \frac{\mathrm{d}x_2}{\mathrm{d}t} = 0. \tag{5.4}$$

Equation (5.3) gives $\mathrm{d}x_1/\mathrm{d}t = (T - T_{\mathrm{G}})/c$ and the first derivative of (5.2) gives $\mathrm{d}x_2/\mathrm{d}t = (\mathrm{d}T/\mathrm{d}t)/k$, and so (5.4) becomes

$$\frac{T - T_{\mathrm{G}}}{c} + \frac{1}{k}\frac{\mathrm{d}T}{\mathrm{d}t} = 0 \tag{5.5}$$

or

$$\frac{\mathrm{d}T(t)}{\mathrm{d}t} + \frac{T(t)}{\tau} = \frac{T_{\mathrm{G}}(t)}{\tau}, \tag{5.6}$$

where $\tau = c/k$ is the relaxation time. $T(t)$ is a function of t driven by $T_{\mathrm{G}}(t)$.

The general solution to this with $T_G(t) = \alpha$ (a constant) for $t > 0$ and 0 for $t < 0$, with tension $T(0)$ at $t = 0$, is

$$T(t) = \alpha + (T(0) - \alpha)\exp(-t/\tau) = T(0)\exp(-t/\tau) + \alpha(1 - \exp(-t/\tau)). \quad (5.7)$$

This can be proved by substituting this in (5.6) and checking the solution at $t = 0$. (See Appendix C.)

Before the tension generator turns on, $T_G = 0$ and $T(0) = 0$. At $t = 0$, $T_G(t) = T_0(= \alpha)$ for a period of time, during which time (5.7) is

$$T(t) = T_0(1 - \exp(-t/\tau)), \quad (5.8)$$

as shown in Fig. 5.16a.

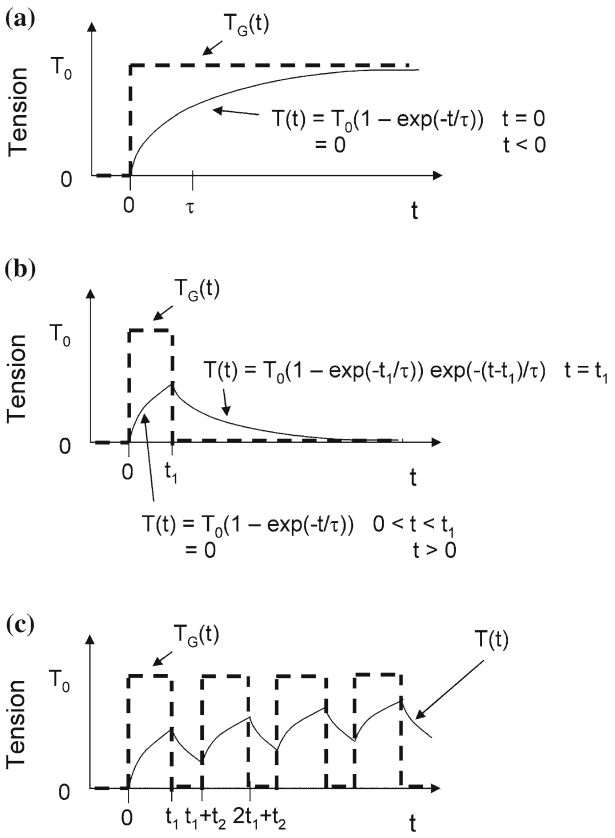


Fig. 5.16 Solutions of the active state muscle model in Fig. 5.15 for the tension with excitation by **a** a step function, **b** a square pulse, and **c** a square wave. The text derives the solution after two sequential square pulses at time $2t_1 + t_2$ in (**c**). (Based on [53])

If the tension generator turns off at time t_1, so $T_G(t) = 0$ for $t > t_1$, then the existing tension at t_1 is

$$T(t_1) = T_0(1 - \exp(-t_1/\tau)), \tag{5.9}$$

and (5.7) (now with $\alpha = 0$) shows that the tension then decays at a rate $\exp(-\Delta t/\tau)$, where $\Delta t = t - t_1$. Therefore, for $t > t_1$

$$T(t) = T_0(1 - \exp(-t_1/\tau)) \exp(-(t - t_1)/\tau), \tag{5.10}$$

as is seen in Fig. 5.16b.

This is the model of a single twitch. We can consider sequences of two twitches separated in time one right after the other (see Fig. 5.16c). Say the activation is off for a time t_2 and is on again at $t_1 + t_2$ for a time t_1 (until $2t_1 + t_2$). Then the tension at the start of the second twitch is

$$T(t_1 + t_2) = T_0(1 - \exp(-t_1/\tau)) \exp(-t_2/\tau). \tag{5.11}$$

For $t_1 + t_2 < t < 2t_1 + t_2$, the tension evolves as given by (5.7) with (5.11) used for the initial tension, $\alpha = T_0$, with t replaced by $t - (t_1 + t_2)$ on the right hand side of the equation (which is a new definition of the starting time). Therefore,

$$\begin{aligned} T(t) &= T_0\{1 + [\exp(-t_2/\tau) - \exp(-(t_1 + t_2)/\tau) - 1] \exp(-(t - (t_1 + t_2))/\tau)\} && (5.12)\\ &= T_0\{1 - \exp(-t/\tau) + \exp(-(t - t_1)/\tau) - \exp(-(t - (t_1 + t_2))/\tau)\}. && (5.13) \end{aligned}$$

At the end of the second twitch $t = 2t_1 + t_2$, so

$$\begin{aligned} T(t) &= T_0\{1 - \exp(-(2t_1 + t_2)/\tau) + \exp(-(t_1 + t_2)/\tau) - \exp(-t_1/\tau)\} && (5.14)\\ &= T_0[1 - \exp(-t_1/\tau)][1 + \exp(-(t_1 + t_2)/\tau)], && (5.15) \end{aligned}$$

which is $(1 + \exp(-(t_1 + t_2)/\tau))\times$ larger than at the end of the first twitch.

At the end of N such twitches, the tension is larger than at the end of the first twitch by a factor $1 + \exp(-(t_1 + t_2)/\tau) + \exp(-2(t_1 + t_2)/\tau) + \ldots + \exp(-N(t_1 + t_2)/\tau)$, which for large N approaches $1/[1 - \exp(-(t_1 + t_2)/\tau)]$. (The geometric sum $1 + x + x^2 + x^3 + \ldots \to 1/(1 - x)$ for $0 < x < 1$.) Therefore the total developed tension in the tetanized state is

$$T_{\text{tetanized}} = T_0 \frac{1 - \exp(-t_1/\tau)}{1 - \exp(-(t_1 + t_2)/\tau)}. \tag{5.16}$$

If the pulses come right after each other, so $t_2 = 0$, this equation becomes $T_{\text{tetanized}} = T_0$, which makes sense.

Figure 5.17 shows a more complete mechanical model of a muscle, with individual sarcomeres modeled.

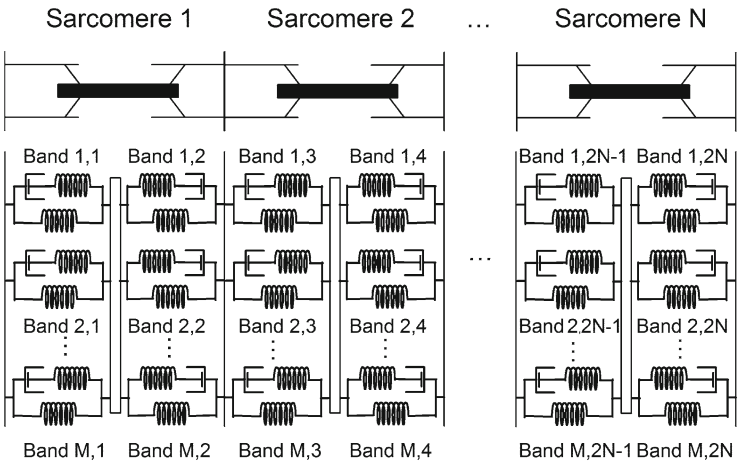


Fig. 5.17 Schematic diagram of a muscle fiber that consists of the simpler models. This more complete muscle model is built from a distributed network of N sarcomeres. (Based on [15] and [61])

5.4 Passive Muscles

A passive or resting muscle with no electrical stimulation has mechanical properties that need to be understood as part of exploring the active state of the muscle because they affect muscle performance during activation. Experimentally it is found that passive muscles are non-Hookean, as in (4.22). Equation (4.27) showed that for larger strains, stress and strain are related by

$$\sigma = \mu' \exp(\alpha L/L_0) - \mu, \tag{5.17}$$

with $\lambda = L/L_0 = \epsilon + 1$ being the Lagrangian strain, where L is the muscle length and L_0 is the relaxed length.

Using this, the force across a passive muscle is related to stress by

$$F_{\mathrm{M}} = \sigma(\mathrm{PCA}). \tag{5.18}$$

Consequently, the force needed to maintain a passive muscle at a length L is

$$F_{\mathrm{M}} = (\mathrm{PCA})\mu' \exp(\alpha L/L_0) - (\mathrm{PCA})\mu. \tag{5.19}$$

For large strains, more general relations need to be used for this neo-Hookean regime and for defining strains. Chapter 4 and Problems 4.20–4.22 explain this further.

5.5 Active/Tetanized Muscles: Microscopic View

Unlike passive organ parts whose lengths change because of applied stresses, the lengths of muscles change and force is generated by the sliding of the thick filaments on the thin filaments, as seen in Figs. 5.18 and 5.19. Because the basic operating unit undergoing this motion is the sarcomere, which is $\sim 2\,\mu$m long, these figures describe a microscopic view of muscle action.

The maximum tension force that can be developed by a muscle depends on several quantities, including the current muscle length, the current rate of muscle contraction, and the physiological cross-sectional area (PCA) of the muscle. Under optimal conditions this force is $\sim 30\,\text{N/cm}^2$ for many muscles in most mammals.

The tension generated by a muscle when the two ends of the muscle are held fixed, called *isometric* conditions, is shown in Fig. 5.19. The maximum force generated by a muscle occurs at the optimal length L_0. (This occurs at a muscle length slightly longer than the initial or resting length L_i, prior to activation.) This maximum level of force decreases to half of this value when the muscle length is 70% (shortened muscle) or 130% (lengthened muscle) of this optimal length. It is almost zero at 170% of this resting length. These observations refer to both the total length of the actual muscle and the length of each $\sim 2\,\mu$m long sarcomere [12].

The physical basis of this tension is the extent of overlap of myosin and actin molecules on the thick and thin filaments. At the resting length the number of myosin/actin pair overlaps is at a maximum, and this number of overlapping pairs decreases when

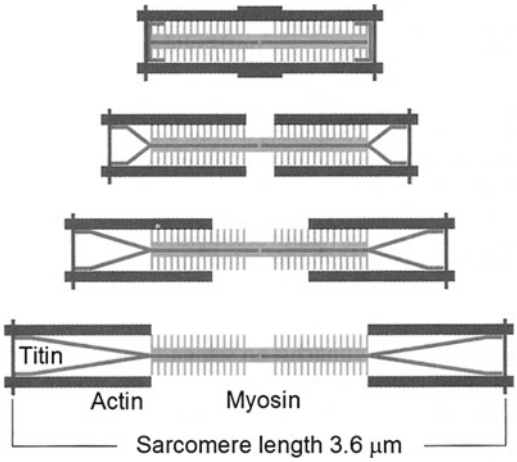


Fig. 5.18 Schematic of the sliding filament mechanism in sarcomeres, with relative sliding of the thick and thin filaments, for sarcomeres of increasing length: 1.6, 2.2, 2.9, and 3.6 μm (which roughly correspond to the sarcomeres highlighted in Fig. 5.19). Note that the elastic material titin is schematically shown to connect the thick filament to the structure; it plays an important role in the elastic properties of muscles. Also see Fig. 5.35. (From [68])

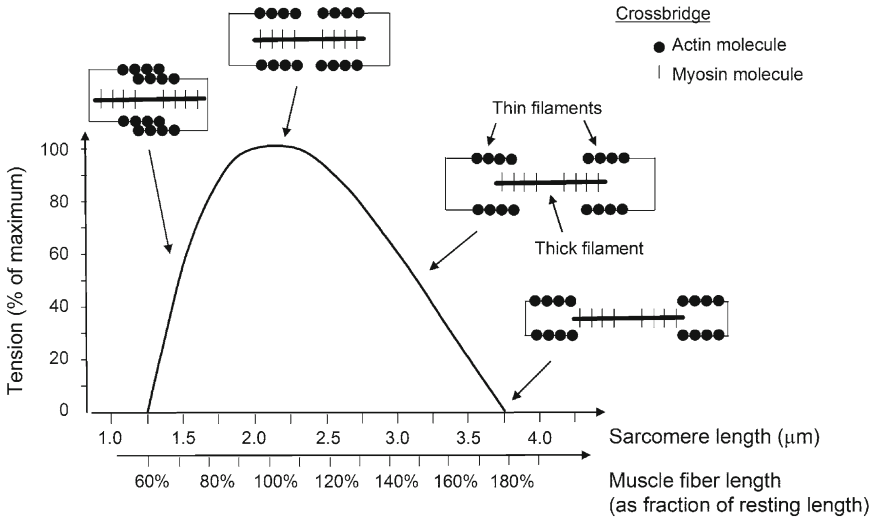


Fig. 5.19 Force versus sarcomere or muscle length, with schematics of the variation of the overlap of the thick (myosin) and thin (actin) filaments for different sarcomere lengths. The sarcomere and total muscle lengths scale the same way. The crossbridge between an actin and myosin molecule is also shown. (Based on [26, 53, 67, 71])

the sarcomere gets shorter or longer. These interactions of these pairs of molecules are known as crossbridges between the thick and thin filaments.

One consequence of changing muscle length is the concomitant changing of joint angle (Problem 5.16). Consequently, the generated force and resulting torque also depend on the joint angle. Figure 5.20 shows the force versus joint angle for several joints.

5.5.1 Total Muscle Tension

Figure 5.21 shows the total tetanized tension versus muscle length. It clearly looks different than the response shown in Fig. 5.19. Also shown in Fig. 5.21 is the passive tension versus muscle length. The contribution of the passive properties of muscle tissue cannot be ignored in modeling activated muscles. The passive component is given by (5.19), and it and the developed tension are added in Fig. 5.21 to give the total tension. In parallel, fusiform muscles, such as the sartorius muscle in the thigh, there are long muscle fibers with relatively little passive material. The maximum seen in the active part is still seen in the total muscle response (Fig. 5.21b). In pinnate muscles, such as the gastrocnemius muscle, there are short fibers and much connective tissue and so the passive material contribution is large and the maximum attributable to the developed portion alone is not seen (Fig. 5.21a).

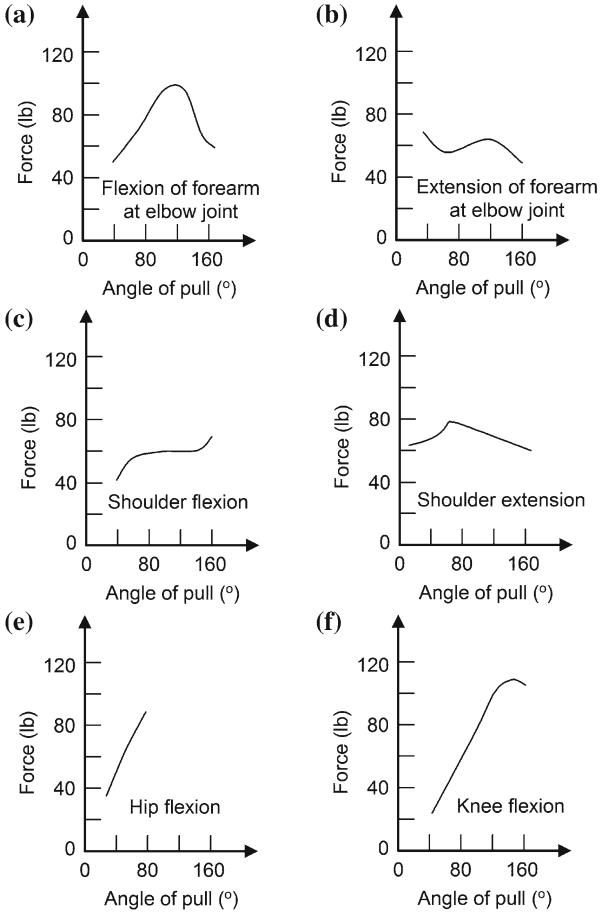


Fig. 5.20 Force versus angle for elbow and shoulder flexion and extension, and hip and knee flexion. Only some of these plots look like the classic force versus length locus (Fig. 5.19, converted to a plot versus joint angle) for a single muscle fiber or sarcomere. (Based on [20])

5.5.2 *Everyday Proof of the Limited Range of Useful Muscle Length*

The decrease in the tension that muscle is able to develop when it is very short or long can be seen in simple demonstrations [4].

For most of us, it is hard to do pull ups (chin ups) or push ups with our arms fully extended, because in this beginning position the muscles we need to use are much longer than their resting length. It is much easier to start these exercises half-way through the motion, with the muscles nearer their resting length.

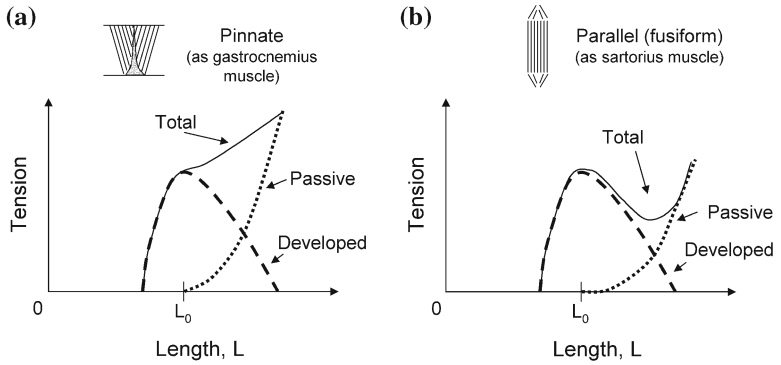


Fig. 5.21 Schematic of passive, developed (active), and total tension versus length for pinnate and parallel muscles. (Based on [5] and [53]. Also see [1])

Grip a ball (or pen) tightly in one fist. Have someone of comparable physical dimensions try to grasp the ball from your fist. Hopefully (and quite likely) he or she cannot. Now use your other hand to push the top of your clenched fist down at the wrist, to flex your wrist downward as much as possible without inflicting undue pain. Now have your colleague try to grasp the ball. It should be a relatively simple task. Why? Before you usually make a fist the relevant muscles are initially near their resting length, but when your wrist is flexed, the relevant muscles are much too long to generate much tension. What is the exact origin of this?

There are 20 muscles in the hand, with a total mass of about 91 g. The total hand itself has a mass of about 600 g. There are 20 muscles in the forearm, with a mass of about 766 g. Nine of these, extensor and flexor muscles with a combined mass of 401 g, are connected to the hand via long tendons (Figs. 5.22 and 5.23). If these were physically in the hand, it would be very bulky.

Now back to holding the ball in a clenched fist. When the wrist is flexed (Fig. 5.24a), with no ball in place, the digital extensor muscles are already stretched, while the digital flexors are slackened (closed circles in Fig. 5.25 a, b). When the wrist is in this position and you try to clench an object, like a ball, the extensors need to be stretched even more and the flexors need to be slackened even more (arrows in Fig. 5.25 a, b). Both muscles enter regimes beyond the normal range of muscle length.

This type of knowledge does have practical applications! You can make an assassin with a knife in hand drop the knife by forcibly bending his or her wrist.

Similarly, you cannot straighten your fingers when your wrist is first hyperextended (hand back the other way). When the wrist is hyperextended (Fig. 5.24b), with no ball in place, the digital extensor muscles are slackened, while the digital flexors are stretched (closed circles in Fig. 5.25 c, d). When the wrist is in this position and you try to straighten your fingers, the extensors need to be slackened even more and the flexors need to be stretched even more (arrows in Fig. 5.25 c, d). Again, both muscles enter regimes beyond the normal range of muscle length.

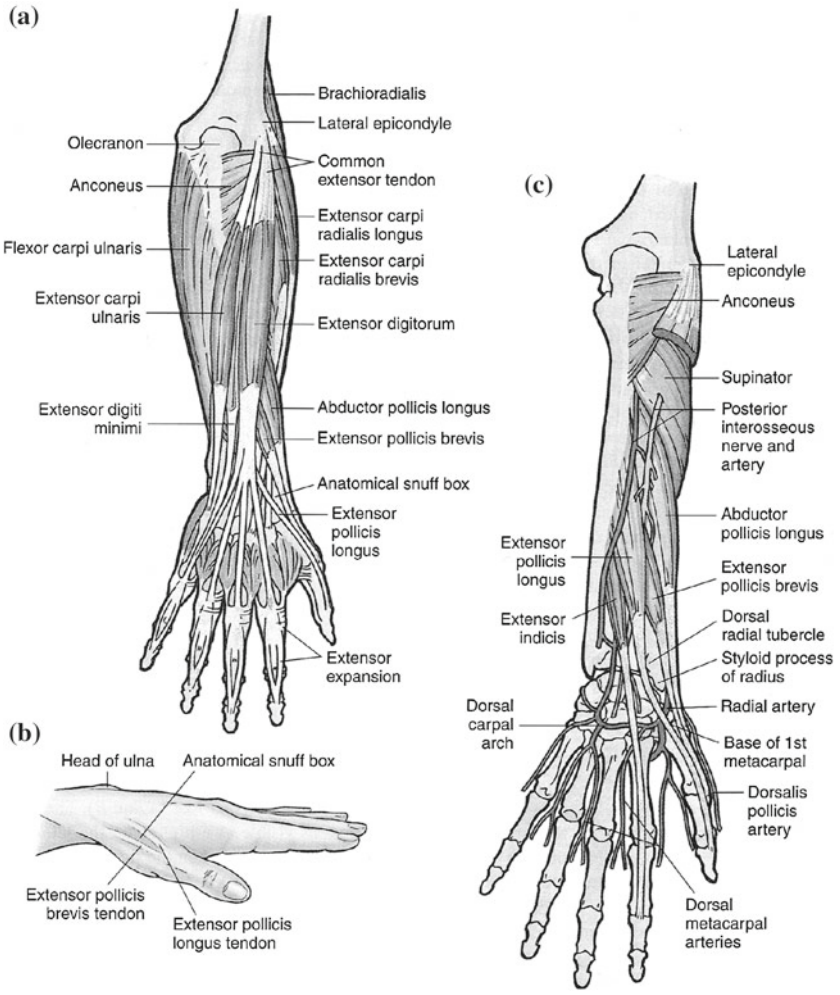


Fig. 5.22 Extensor muscles of the forearm, with **a** posterior view of superficial muscles, **b** selected features of the hand, and **c** deeper muscles, along with selected arteries. Tendons extending to the hand are also seen. (From [57]. Used with permission)

5.6 Hill Force–Velocity Curve

The maximum tension T a muscle can develop also depends on how fast it contracts. An equivalent way of saying this is the speed at which a muscle can shorten depends on how much tension it must generate to overcome a load, such as a mass hanging at the end of it. The contraction speed is $v = -\mathrm{d}L/\mathrm{d}t = |\mathrm{d}L/\mathrm{d}t|$, and it is measured with the muscle fixed at one end and with a fixed load on the other, i.e., isotonically. (Again, this fixed load can be a hanging mass.) The maximum tension and contraction

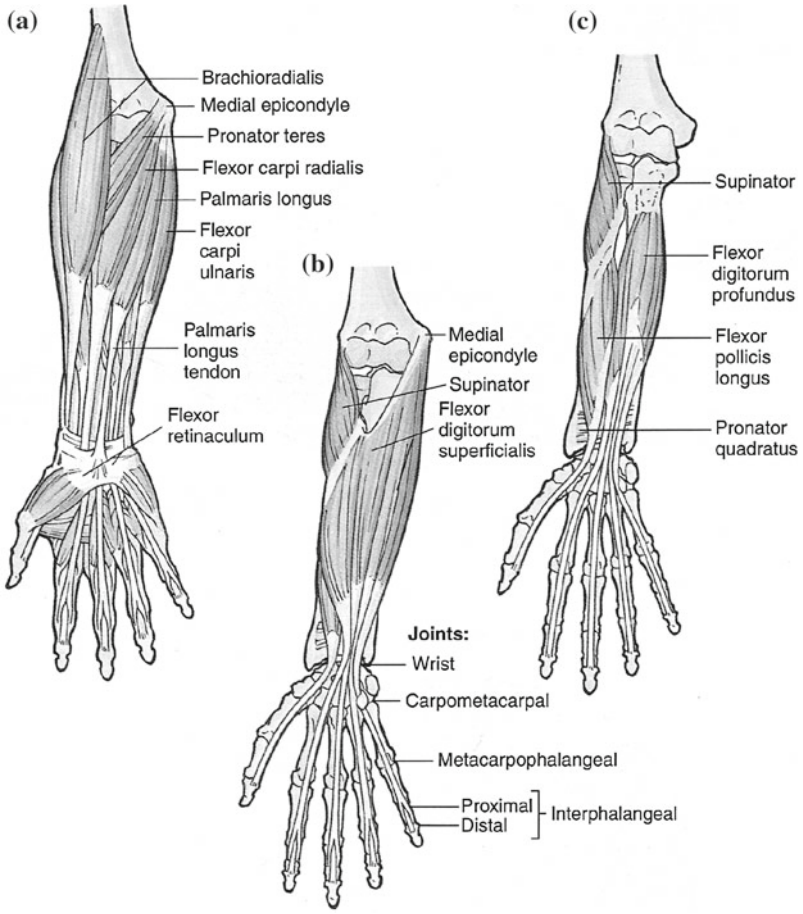


Fig. 5.23 Flexor muscles of the forearm, with anterior views of the first (most superficial) layer in (**a**), the second layer in (**b**), and the third and fourth layers in (**c**). Tendons extending to the hand are also seen. (From [57]. Used with permission)

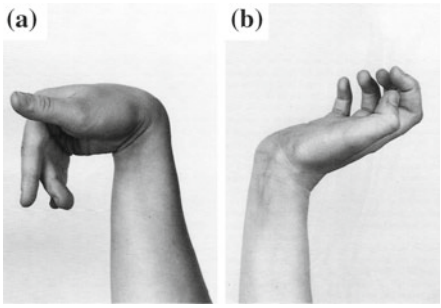


Fig. 5.24 It is difficult to clench your fist with a strongly flexed wrist (**a**) and straighten your fingers with a strongly hyperextended wrist (**b**). (From [4]. Copyright 1992 Columbia University. Reprinted with the permission of the press)

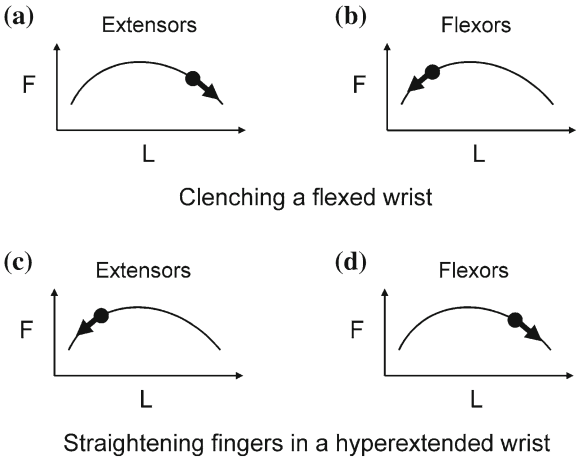


Fig. 5.25 Schematic of forces on extensors and flexors for clenching a flexed wrist and straightening fingers in a hyperextended wrist as shown by the closed circles. The arrows show the continued slackening or stretching described in the text

speed are experimentally observed to follow the hyperbolic dependence in the Hill force–velocity curve (Fig. 5.26) [31, 78], with the muscle shortening faster with a smaller load mass. Note that muscles do work by contracting, and as such the term tension really refers to the tension felt by objects attached to them. (By the way, Archibald Vivian Hill, who formulated this force–velocity relation, shared the Nobel Prize in Physiology or Medicine in 1922 for his discovery relating to the production of heat in the muscle; much of his work concerning the force generated by muscles occurred after his Nobel-Prize-winning research.)

The maximum tension $T_{\max}$ that can be developed occurs for $v = 0$ (isometric conditions), which is the limit of very slow muscle contraction, $T(v = 0) = T_{\max}$ (sometimes called T_0). T decreases and becomes zero at $v_{\max}$. One form of the Hill force–velocity curve is

$$(T(v) + a)(v + b) = (T_{\max} + a)b \tag{5.20}$$

or

$$T(v) = \frac{bT_{\max} - av}{v + b}, \tag{5.21}$$

which is plotted in Fig. 5.26. It is valid from $v = 0$ to $v_{\max}$. The right hand side of (5.20) represents the point on the hyperbola with maximum tension, with $v = 0$ and $T(v = 0) = T_{\max}$. We could have chosen the values at any other point on the hyperbola, such as that with minimum tension with $v = v_{\max}$ and $T(v = v_{\max}) = 0$, for which the right-hand side would have been $a(v_{\max} + b)$. Clearly these right-hand sides are equal, so $(T_{\max} + a)b = a(v_{\max} + b)$; this means $v_{\max} = (b/a)T_{\max}$, so there are only three independent parameters. The effective response time of a muscle of length L is $L/v_{\max}$. (Why?)

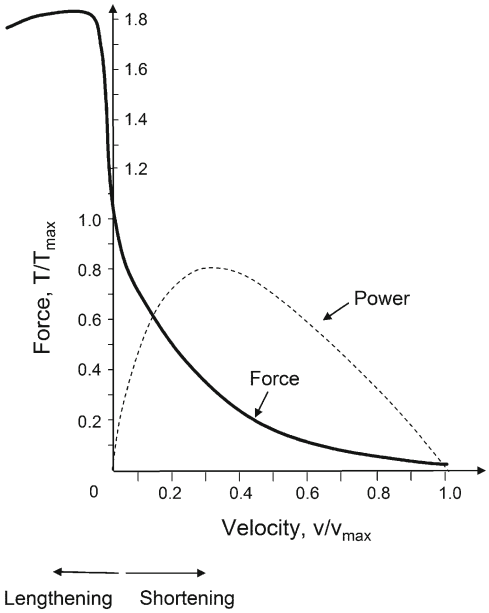


Fig. 5.26 Hill force–velocity curve. Both muscle shortening and lengthening are shown. Power = Force × velocity, is also shown, with arbitrary vertical units. (Based on [53])

After rearranging terms, (5.21) can be expressed as

$$v' = \frac{1 - T'}{1 + T'/k}, \tag{5.22}$$

where $v' = v/v_{\max}$, $T' = T/T_{\max}$, and $k = a/T_{\max} = b/v_{\max}$. Typically, $0.15 < k < 0.25$. Table 5.9 gives the parameters in the force–velocity curve for several muscles. This Hill force–velocity curve can also be expressed in terms of the torque applied to a joint and the angular speed of that joint (see Problem 5.20).

The empirical Hill force–velocity curve can be derived by considering the statistics (more precisely the statistical mechanics) of the crossbridges in the sliding filament model. The derivation of this Huxley model is outlined later in this chapter. Whereas the Hill model works well in steady state, it does not when muscle conditions are changed rapidly (Problem 5.24); the Huxley model works well in both cases.

We saw in (3.87) that the power needed to move an object is Fv, where F is the (constant) force on the object and v is the speed with which it is moved. Therefore the power generated by the muscle is

$$P(v) = T(v)v = v\frac{bT_{\max} - av}{v + b}. \tag{5.23}$$

Problem 5.21 shows that this is a peak when $v \sim 0.3v_{\max}$, for which $T \sim 0.3T_{\max}$. At this speed the power attains a peak value of $\sim 0.1T_{\max}v_{\max}$ (Fig. 5.26). Bicycles have

Table 5.9 Parameters for the Hill force–velocity relationship for skinned muscles, with forces normalized per unit area

Muscle	Isometric force, T_{max} (MPa)	Maximum velocity, v_{max} (FL/s)[a]	a/T_{max}
Human soleus (15°C)[b]	0.145	0.52	0.037
Human gastrocnemius (15°C)[b]	0.136	0.64	0.034
Monkey soleus (15°C)[b]	0.146	0.7	0.044
Monkey gastrocnemius (15°C)[b]	0.160	0.69	0.040
Rabbit psoas (20°C)[c]	0.246	3.26	0.10
Rat soleus (20°C)[c]	0.234	1.94	0.05

[a] FL is the fiber length
[b] [77]
[c] [75]
Using data from [51]

gears that take advantage of this, keeping the muscle shortening velocity near the value where this peak power is achieved. This is analogous to shifting gears in cars traveling at different speeds so their engines can operate at an rpm near maximum torque.

The Hill force-velocity curve actually refers only to the contractile element in the muscle. We can relate the tension across the entire muscle or muscle fiber, T, to its length, L, by using Hill's two-element model for a skeletal muscle [40, 41]. This is the contractile element described by the Hill force–velocity curve in series with an elastic element; this is the model shown in Fig. 5.15 without the dashpot. The tension T is the same across the entire muscle, the contractile element, and the spring. The length of the muscle is the sum of the lengths of the contractile element and spring, $L = x_1^{\mathrm{T}} + x_2^{\mathrm{T}} = x_1^{\mathrm{E}} + x_1 + x_2^{\mathrm{E}} + x_2$ (using the notation in the discussion of Fig. 5.15), and so $\mathrm{d}L/\mathrm{d}t = \mathrm{d}x_1/\mathrm{d}t + \mathrm{d}x_2/\mathrm{d}t$. The contraction speed of the contractile element is $v = -\mathrm{d}x_1^{\mathrm{T}}/\mathrm{d}t = -\mathrm{d}x_1/\mathrm{d}t$, so $\mathrm{d}x_2/\mathrm{d}t = \mathrm{d}L/\mathrm{d}t - \mathrm{d}x_1/\mathrm{d}t = \mathrm{d}L/\mathrm{d}t + v$. Using (5.20), this speed $v = b(T_{max} - T)/(T + a)$. The time rate of change of the tension can be determined using the chain rule, $\mathrm{d}T/\mathrm{d}t = (\mathrm{d}T/\mathrm{d}x_2)(\mathrm{d}x_2/\mathrm{d}t)$. With $\mathrm{d}T/\mathrm{d}x_2 = k$, the spring contant, the tension across the entire muscle T is related to its length L by

$$\frac{\mathrm{d}T}{\mathrm{d}t} = k\left(\frac{\mathrm{d}L}{\mathrm{d}t} + \frac{b(T_{max} - T)}{T + a}\right). \tag{5.24}$$

This equation can only be used if T_{max} does not change (or does not change much) during muscle shortening at constant speed, such as for small contractions near the peak of the tension-length curve (Fig. 5.19).

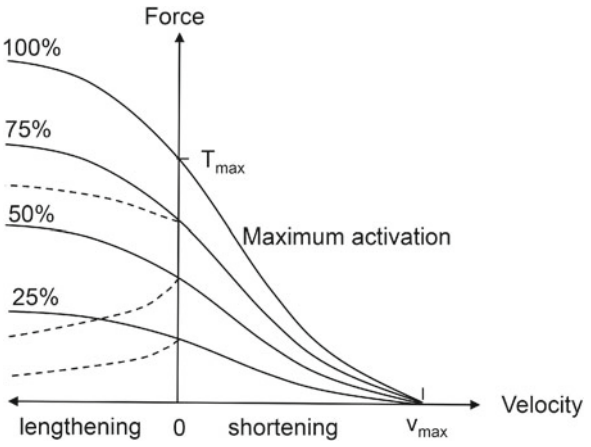


Fig. 5.27 Force–velocity curves at different levels of muscle activation, for a given length muscle. All shortening curves follow the Hill force–velocity curves, but for lengthening the curves are different for isotonic activity (*solid curves*) and isovelocity activity (*dashed curves*). (Based on [81])

At lower levels of muscle excitation, there is less contractile force at each velocity, as is seen in Fig. 5.27. Moreover, muscles usually do not maintain a zero speed, as is presumed in Fig. 5.19, or a constant length, as is presumed in Fig. 5.26. The response of a muscle really depends on both its length and the contraction velocity, as is depicted in Fig. 5.28. This shows that that when the muscle length is not L_0, the maximum contraction speed is less than v_{max}.

Simple models of motion commonly assume that the muscle force depends on muscle length (and not contraction speed, which is implicitly assumed to be slow) or contraction speed (and not muscle length, which is implicitly assumed to remain near that for maximum muscle force), but not both (as in Fig. 5.28). This is done either because it is simpler that way or the specific conditions actually warrant it. (See Problems 5.25 and 5.26.) We used these alternative assumptions in Chap. 3 to model throwing and jumping.

So far we have considered muscle shortening from speeds 0 to v_{max}. This is important in doing positive work. However, Fig. 5.26 is seen to extend to the left of $v = 0$, where there is muscle lengthening ($dL/dt > 0$). The tension that is developed in this eccentric contraction regime is about $1.8T_{max}$. In this regime the muscle is used for braking, not performing positive work. This application is used for braking motion (eccentric), resisting pulling, isometric (constant muscle length) activities, and keeping the body stable—as in standing. Remember that 100 J is lost in each step during running. Almost half of it is absorbed in the braking action of the leg muscles. When the muscles are not fully activated, this very large tension in braking always occurs for isotonic motions but not always for isovelocity modes, as seen in Fig. 5.27.

Figure 5.29 shows experimental data for the torque that can be developed in knee extension versus the speed of angular change [21]. In this figure the generated forces are proportional to this torque and the speed of muscle length change is proportional to this speed of angular change. This is analogous to interrelating muscle force

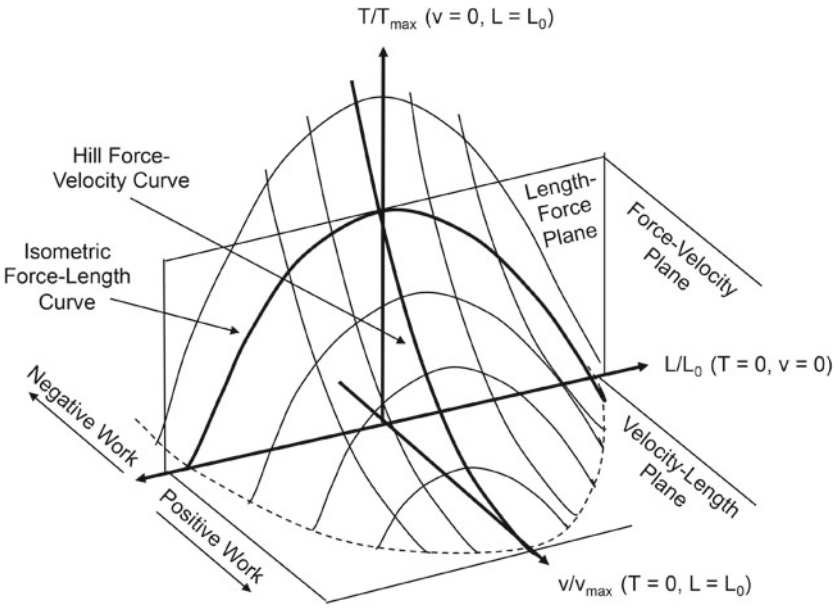


Fig. 5.28 Three-dimensional plot showing the tension of the active part of a muscle versus velocity and length. This is for maximum muscle activation and excludes the passive muscle response. (Based on [81])

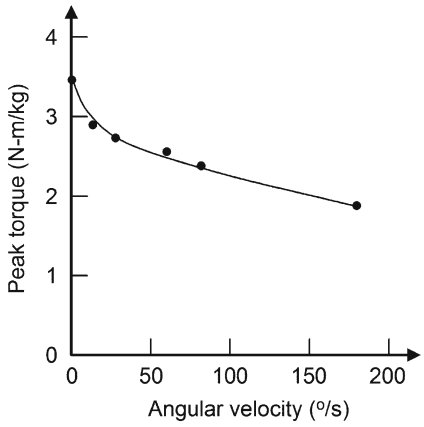


Fig. 5.29 Torque (per unit body mass) versus knee angular velocity for knee extension. (Based on [21] and [73])

vs. length with knee torque vs. angle. (See Problem 5.20.) Fig. 5.12 shows that the maximum angular speed and torque depend on the fraction of FT muscles.

In weightlifting a heavier load can be raised if it is raised quickly, i.e., if it is "jerked," than if it is moved more slowly and smoothly. This is not really contradictory to the Hill force–velocity curve because the lifter is taking advantage of the higher muscle forces that are possible before the filaments can pick up speed. So, you can

lift heavier weights by rapid lifting to shoulder height and then a jerk to full vertical extension—the "clean-and-jerk"—than by rapid lifting to shoulder height and then slow lifting to full vertical extension—the "press." (By the way, the press is no longer a competitive event because it is hard to distinguish from the clean-and-jerk.) A third weightlifting motion is the "snatch," in which the weight is raised in one continuous motion. The world records for the snatch and the clean-and-jerk are 151.0 kg and 190.0 kg, respectively for women ($m_b \geq 75$ kg, as of 2014) and 214.0 kg and 263.5 kg for men ($m_b \geq 105$ kg). (For more on this, see Problem 5.7.)

5.7 The Combined Muscle-Tendon System

The developed force, total length, and contraction speed of the muscle itself and those of the combined muscle-tendon system can be very different. We have just examined the properties of muscles and examined those of tendons in the previous chapter and will now look at the combined system because it is what is actually important. Moreover, in surgical procedures involving tendon transplants, understanding and measuring the passive and active properties of combined muscle-tendon systems can be very helpful [47, 48]; this is much like the need for compatibility of mechanical properties in assessing the suitability of biomedical implants that we saw in Chap. 4.

In the combined system, force can be sustained when the muscle fiber length exceeds $\sim 0.7L_0$, where L_0 is (still) the optimal muscle fiber length, and this occurs when the length of the total muscle + tendon system exceeds $L_{T,s} + 0.7L_0$, where where $L_{T,s}$ is the slack or relaxed length of the tendon [82]. (This is true for fusiform and parallel muscles, and this and other parameters we will define need to be modified for pennate muscles.)

The ratio $L_{T,s}/L_0$ is an important parameter in describing the combined effect of the muscle and tendon. When this is large, say ~ 10, the overall system is very compliant and the tendon affects the overall force versus length curves very strongly, and when it is small, ~ 1, the overall system is considered to be stiff and the effect of the tendon is relatively small. In humans, this ratio is ~ 11 for the soleus, ~ 9 for the gastrocnemius and ~ 7 for other plantarflexor muscles; ~ 3 for the vasti and ~ 5 for the rectus femoris in the quadriceps; ~ 2 for the semitendinosus and ~ 7 for the semimembranosus/biceps (long) in the hamstrings; and ~ 0.2 for the hip uniarticular muscles [82].

Figure 5.30 shows how the response of the muscle alone (including its active and passive parts, as in Fig. 5.21) changes when the tendon is included. What is plotted is the total length less that of the slack tendon, so it is really the muscle length plus the change in tendon length. For the very stiff tendon depicted in (a), the curve is stretched laterally only a bit and the change in the ascending region ($L < L_0$), where the passive response of the muscle is small, is changed only a little. For the very compliant tendon in (b), the response is very different, as is seen by the very large changes in the ascending region. This is very distorted compared to that for the muscle only, with the length of the ascending part of the curve being much longer,

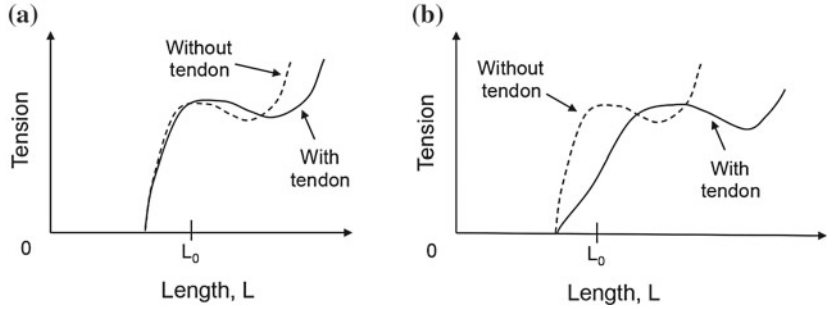


Fig. 5.30 Force versus length for a combined muscle and tendon, including the maximum activated response and the passive response of muscles for tendons that are either **a** very stiff ($L_{T,s}/L_0 = 3$) or **b** very compliant $L_{T,s}/L_0 = 15$ for isometric conditions. (Based on [82])

and the maximum of the developed muscle occuring closer to $1.5L_0$ than to L_0; at the peak developed force the tendons are stretched a large fraction of the muscle optimal length.

So far we have assumed isometric conditions. The relative length of the tendons to muscle also affects the contraction speed and dynamics of the total system. For very large $L_{T,s}/L_0$, as with the soleus, the rate limiting features of the generation of force are the dynamics of the muscle-tendons system; for very small ratios, such as for the uniarticular hip muscles, the activation of the muscle is the rate-limiting feature; and for very intermediate ratios, as for the rectus femoris, both processes are important [82].

5.8 The Sliding Filament Model: Nanoscopic View

The body has several motor proteins that induce directed movement. Myosin (Figs. 5.7, 5.18, 5.19, and 5.31) does it in muscles [62]. Kinesin (kai-nee'sin) and dynein (die-nee-in) direct transport on microtubules in the body (Fig. 5.32) [6, 10, 11, 24, 27, 42, 69, 74]. (Both are used in long-range transport in the body. Within the cell, kinesin is used for transport from the center of the cell to its edge, while dynein is used for transport from the edge of the cell to its center.) Such direct transport is preferred to diffusion because it provides a directed motion and a motion that is faster than diffusion. As is described in more detail in Chap. 7, in one dimension diffusion leads to a slow gaussian-like, undirected spreading of the species over a distance $x \sim \sqrt{2D_{\text{diff}}t}$ in a time t, where D_{diff} is the diffusion coefficient.

Directed motion is necessary for muscle movement. Fig. 5.31 shows the head of a myosin molecule from a thick filament attached to an actin molecule labeled x on a thin filament (Fig. 5.31a). This single coupling is a crossbridge. This stage in the interaction is called the cocked position. ATP is hydrolyzed to form ADP + inorganic phosphate + energy (see Chap. 6) and this energy is used to contort

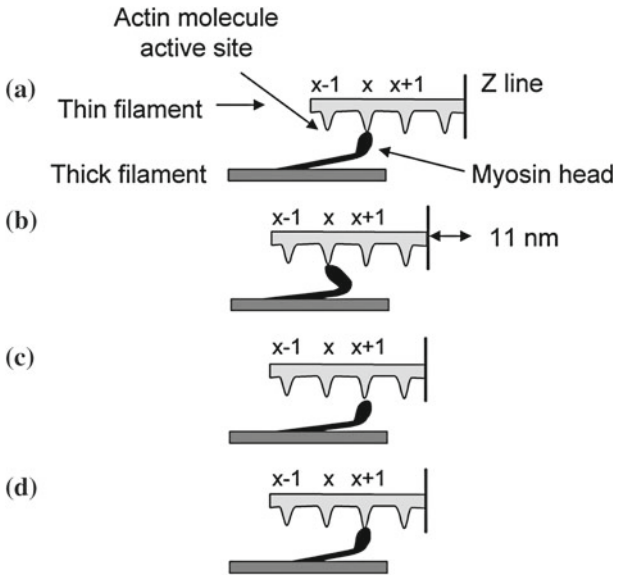


Fig. 5.31 Sliding of myofilaments, with **a** binding of the myosin head to an active actin site (site x), **b** the power stroke in which the myosin head hydrolyzes ATP and distorts, and moves the thin filament, **c** the myosin head detaches and relaxes, and **d** the myosin head reattaches to a new actin site ($x + 1$) for a new cycle. (Based on [67] and [71])

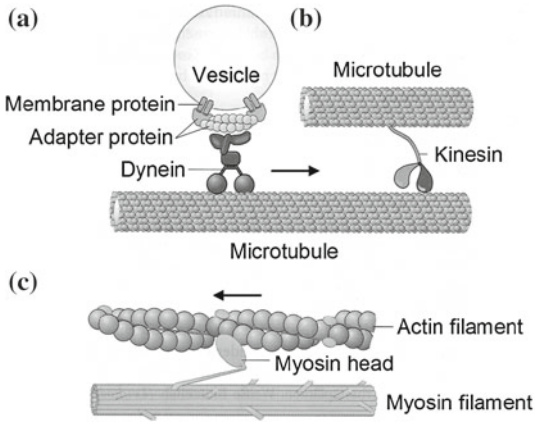


Fig. 5.32 Comparison of motor proteins, with **a** transport of a vesicle along a microtubule track by dynein, **b** transport of a microtubule on a microtubule track by kinesin, and **c** motion of the actin filament by a myosin filament. (From [68])

the myosin molecule, which is still bound to the same actin molecule, to a new configuration. This motion, along with the simultaneous action of other crossbridges on the same filaments, causes the thin filament to slide to the left. This is the power stroke stage. The myosin molecule detaches (the detach phase), relaxes, and then binds to the adjacent actin molecule labeled $x + 1$ (Fig. 5.31d) on the thin filament

(the bind phase). *How many crossbridges are there in a muscle? How much force is generated per crossbridge?* We need to connect this nanoscopic crossbridge view with the macroscopic observation that muscles generate up to $\sim 30\,\mathrm{N/cm^2}$.

Each muscle has 10^4–10^6 muscle fibers. Each fiber is 10–80 μm in diameter. Let us say 50 μm is typical so there are $\sim 1/(50\,\mu\mathrm{m})^2 = 4 \times 10^4$ fibers/cm^2. Each fiber has several hundred to several thousand myofibrils, and let us say there are 2,000 of them. Each myofibril has about 1,500 myosin thick filaments and 3,000 actin thin filaments, and $\sim 1 \times 10^4$–1.7×10^5 sarcomeres. These filaments are arranged in an ordered manner as in Fig. 5.6a, b (lower). Each thick filament has 200 myosin molecules.

In a given sarcomere, all the crossbridges are in parallel, so the forces of each add for each filament. This is analogous to many people pulling on a rope in a tug-of-war or rowing together in a crew boat or a galley ship, where their combined forces add. All filaments in a sarcomere (myofibril) add in parallel and so these forces add. All myofibrils in a fiber add in parallel, so their forces add. All muscle fibers add in parallel and so their forces add.

Therefore, within the width of a half sarcomere there are: 4×10^4 fibers/cm^2 × 2,000 myofibrils/fiber × 1,500 thick filaments/myofibril × 200 myosin molecules/thick filament $\simeq 1 \times 10^{13}$ myosin molecules/cm^2. This means that the force developed across in each crossbridge is

$$\sim \frac{30\,\mathrm{N/cm^2}}{1 \times 10^{13}\,\text{myosin molecules/cm}^2} \sim 3 \times 10^{-12}\,\mathrm{N} = 3\,\mathrm{pN}. \tag{5.25}$$

This is also a conclusion of Huxley's sliding filament model of the Hill force–velocity curve. Direct measurement of the interaction of a single actin–myosin crossbridge using optical tweezers (which are focused laser traps used to confine these molecules) has shown that 3–4 pN is generated per crossbridge and the power stroke distance (Fig. 5.31b)—the relative motion of the filaments per ATP hydrolysis—is 11 nm (Fig. 5.33). This is truly a nanoscopic view of muscles. (In Problem 5.30 you can calculate the work by this motion and characterize it in terms of the fraction of the energy released in the hydrolysis of ATP, which is described in the next chapter.)

In examining the effect of the crossbridges in sarcomeres in exerting forces, we considered only the crossbridges in the sarcomeres in a cross-section of the muscle. These add to give the total force/area for the muscle. However, there are $\sim 1 \times 10^4$–1.7×10^5 sarcomeres along the length of each fiber in the muscle. These tens of thousands of sarcomeres do not increase the force exerted by the muscle, but they serve a quite important, though very different, function. The necessary amount of rotation about joints can only occur if there is enough of a change in the muscle length (or tendon plus muscle length). Because the effective change in sarcomere and muscle length is only about ±30%, the change in total length is controlled by the resting muscle length, which depends on the number of sarcomeres along the length of the fibers. (This is related to the second advantage of the pinnated design of muscles.)

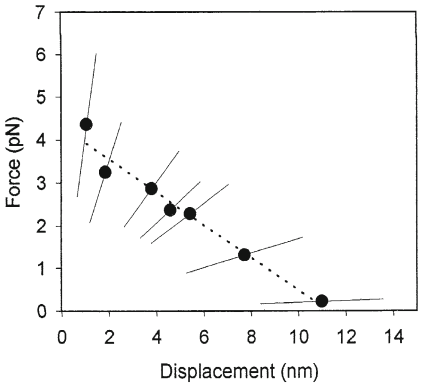


Fig. 5.33 Force versus displacement curve of a single myosin molecule interacting with an actin molecule during a powerstroke, as measured by "optical tweezers." (From [17]. Used with permission of the Biophysical Society)

Our crossbridge thick/thin filament model of the sarcomere is fine so far, but real life is a bit more complicated. For example, it is geometrically more complex. Two actin filaments are actually intertwined, producing a double helix and a tropomyosin strand runs down the grooves of this structure. (Tropomyosin is a regulatory protein, and this strand is different from the thick myosin strand.) Six actin filaments surround each myosin filament, and each actin filament is surrounded by three myosin filaments (Fig. 5.6). Moreover, this attachment/detachment cycle does not happen unless there are Ca^{2+} ions present to induce a topological change that causes the myosin head and actin to be physically close. These ions are present for a limited time after a nerve impulse arrives at the muscle. Also, for force to be exerted for useful motion, there must be some anchoring point and mechanism, and this is provided by the anchoring of each actin filament to a transverse structure by a titin molecule.

Furthermore, the model of concerted motion in which billions or trillions of myosin molecules all attach to actin, cock, then detach, and reattach in perfect unison seems highly improbable [33, 34]. It appears that instead of this concerted and rapid sequence of pull/wait, pull/wait, ... cycles, there are myosin molecules in different parts of the cycle at a given time—but all forming crossbridges pulling in the same direction. (This is like a boat with four oarsmen and eight oars, with, say only two oars in the water and propelling the boat at any one time.) There is a cycle time, τ_{cycle}, for a given crossbridge, during which time the myosin is attached to the actin for a time τ_{on} and is detached and inactive for a time τ_{off}, with $\tau_{\text{cycle}} = \tau_{\text{on}} + \tau_{\text{off}}$. During the on-time the crossbridge moves the attached phase via hydrolysis by the powerstroke distance or working distance δ, which is about 5.3 nm in vitro for skeletal muscle myosin II. (We will approximate this as 5 nm.) The duty ratio r_{duty} is the fraction of time that the crossbridge is attached:

$$r_{\text{duty}} = \frac{\tau_{\text{on}}}{\tau_{\text{on}} + \tau_{\text{off}}} = \frac{\tau_{\text{on}}}{\tau_{\text{cycle}}}. \tag{5.26}$$

For myosin τ_{on} is $\sim 10^{-3}$ s and r_{duty} is ~ 0.02, so a given myosin molecule is in contact with an actin filament for a relatively small fraction of the time.

For the motion of the strands to be continuous there have to be a minimum of N_{min} heads at different phases during the cycle time τ_{cycle}, where

$$N_{min} = \frac{1}{r_{duty}}. \tag{5.27}$$

With $N_{min} \sim 1/0.02 = 50$, only about six of the myosin heads of the roughly 300 myosin crossbridges in a half a filament are attached and exerting a force at any given time.

During a complete crossbridge cycle, which is the time it takes to complete one ATP-hydrolysis cycle, the motor moves a distance Δ, where

$$\Delta = \frac{\delta}{r_{duty}}. \tag{5.28}$$

This is typically $\sim 5\,\text{nm} \times 50 = 250\,\text{nm}$ (for low loads).

The cycle time is related to the rate constant for the hydrolysis of an ATP molecule k_{ATPase}

$$\tau_{cycle} = \frac{1}{k_{ATPase}}. \tag{5.29}$$

Using $k_{ATPase} \sim 20$/s, we see that $\tau_{cycle} \sim 0.05$ s and this is consistent with $\tau_{cycle} = \tau_{on}/r_{duty} = 10^{-3}$ s/0.02. The relative speed of the sliding actin and myosin filaments is given by the total displacement during the cycle time v, which is

$$v = k_{ATPase}\Delta = \frac{\Delta}{\tau_{cycle}} = \frac{\delta}{\tau_{on}}. \tag{5.30}$$

So we see that $v = 5\,\text{nm}/10^{-3}\,\text{s} = 5\,\mu\text{m/s}$.

In addition to δ and Δ, there is a third characteristic distance, d, which is the path distance or step size between consecutive binding sites, and is ~ 36 nm for this crossbridge. This distance is related to Δ by $\Delta = nd$, where n is an integer. In the schematic of Fig. 5.34, $\delta \ll d$ and $\Delta = 3d$, and so $n = 3$. In the simpler model of concerted motion of actin–myosin crossbridges, it was implicitly assumed that these three characteristic distances were the same, and that r was almost 1. In contrast to the real $\ll 1$ duty cycle of myosin, in transport by conventional kinesin the motor protein is just about always in contact with the tubule (and r is almost 1), with δ being about 8 nm, $d = 8$ nm, and $\Delta = 16$ nm.

What is the maximum mechanical power provided by the legs during exercises like rapid step climbing? How is this provided by these crossbridges? Let us assume that the calf and thigh muscles both contract a distance of 3 cm during a step and calculate the energy needed to perform those needed contractions during a step time of 0.25 s, and therefore the necessary power. If these muscles both have an

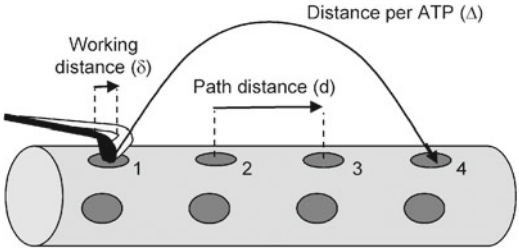


Fig. 5.34 Distances associated with a more sophisticated model of the actin–myosin crossbridge. This crossbridge moves parallel to the filament axis and skips two potential binding sites during a hydrolysis cycle. The duty ratio is low in this example. (Based on [34])

average total diameter of 10 cm and the effective diameter of a myosin filament is 20 nm, the number of myosin filaments in a cross-section of each muscle is about $(10\,\text{cm})^2/(20\,\text{nm})^2 = 2.5 \times 10^{13}$, and so there are about 5.0×10^{13} filaments in cross-section for both muscles. There are six chains of side chains per myosin filament (as seen in Fig. 5.6), so there are 3.0×10^{14} crossbridges in cross-section.

If the attachment/detachment of each crossbridge contracts the chain by 5.3 nm, because each muscle contracts by 3 cm the total number of crossbridge attachments/detachments during the motion is roughly (3.0×10^{14}) $(3\,\text{cm}/5\,\text{nm}) = 1.8 \times 10^{21}$. If each crossbridge motion requires the hydrolysis of one ATP molecule, which releases 14 kcal/mole $= 9.7 \times 10^{-20}$ J (as discussed in Chap. 6), the amount of energy needed is $\sim(1.8 \times 10^{21})(9.7 \times 10^{-20})\,\text{J} = 175\,\text{J}$. If this energy is used in 0.25 s, the power is 175 J/0.25 s = 700 W. This is consistent with the metabolic powers needed for vigorous activities such as step climbing (Problem 5.39 and Chap. 6). Note that part of the uncertainty in the final answer is the very rough estimation of the effective cross-sections of the muscles used.

5.8.1 Statistical Model of the Hill Force–Velocity Curve (Advanced Topic)

A statistical model of the crossbridges in the sliding filament model reproduces the Hill force–velocity curve and agrees with experimental studies better than that curve (when adapted for muscle fibers). We will overview the derivation of this Huxley model here and in the problems; detailed analysis is beyond the level of this text and can be found in [19, 32, 36–38, 40, 41, 53]. This model includes the key points but not all the details of the sliding filament model.

Consider the crossbridge in Fig. 5.31 [41]. Let us focus on how the myrosin head interacts with only one actin molecule, say the one labeled x, but now let x refer to the coordinate of this actin molecule, referenced to the myosin head (laterally, along the x axis) for a relaxed configuration. During sliding the value of x decreases

and the contraction speed $v = -\mathrm{d}x/\mathrm{d}t$ (or $-\partial x/\partial t$). Let us call the fraction of actin molecules with displacement x with bound crossbridges $n(x)$. A new bound crossbridge can form only if there is no existing crossbridge (which has probability $1-n(x)$) and this can occur at a rate $f(x)$. A bound crossbridge can detach if it exists (which has probability $n(x)$) and this can occur at a rate $g(x)$. So, the total change of crossbridge density with time, $\mathrm{d}n/\mathrm{d}t$, is $(1-n(x))f(x) - n(x)g(x)$. This is due to changes that depend on time explicitly and changes that depend on time implicitly due to the sliding. Using the chain rule, this is $\mathrm{d}n/\mathrm{d}t = \partial n/\partial t + (\partial n/\partial x)(\partial x/\partial t)$, and combining these components leads to the Huxley model

$$\frac{\partial n(x,t)}{\partial t} - v(t)\frac{\partial n}{\partial x} = (1-n)f(x) - ng(x). \tag{5.31}$$

If the restoring force for a given displacement x is $r(x)$ ($= kx$ for a spring model), the tension (or force) exerted by the muscle is

$$T = N\int_{-\infty}^{\infty} r(x)n(x,t)\mathrm{d}x, \tag{5.32}$$

where N is the total number of crossbridges (bound and unbound).

So far the solution is valid for any time variation. For steady-state conditions with the filament sliding at constant speed v and $\partial n/\partial t = 0$, as should be presumed in using the Hill force-velocity curve, (5.31) reduces to

$$-v\frac{\mathrm{d}n(x)}{\mathrm{d}x} = (1-n(x))f(x) - n(x)g(x). \tag{5.33}$$

For very slow speeds (isometric conditions), the left hand side approaches 0, and the maximum tension develops, and using (5.32):

$$T_{\max} = N\int_{-\infty}^{\infty} r(x)\frac{f(x)}{f(x)+g(x)}\mathrm{d}x. \tag{5.34}$$

This solution can produce the Hill force-velocity curve, but it does not have the deficiencies associated with using that model. A model with the rate functions chosen that can be solved analytically is analyzed in Problems 5.32–5.37. More sophisticated choices of $f(x)$ and $g(x)$ can be used to model muscles using the Huxley-like model as the contractile element in the Hill fiber muscle model [41].

This Huxley model assumes each myosin head binds to a given actin site, because the other sites are far away. In an improved, more generalized binding model [41] the myosin can bind to any actin site, which are separated on the thin filament by Δx, by modfying (5.31) to arrive at:

$$\frac{\partial n(x,t)}{\partial t} - v(t)\frac{\partial n}{\partial x} = (1 - \sum_{j=-\infty}^{\infty} n(x+j\Delta x,t))f(x) - ng(x). \tag{5.35}$$

The sum can be be converted into an integral to obtain the simpler continuous binding site model.

5.9 Titin

The titin molecule is the third filament in the sarcomere, in addition to the thick and thin filaments, and is responsible for the elastic and viscoelastic response of the sarcomeres [44–46, 49, 56]. In ways it is as important to the function of muscles as are the two active filaments. Titin is a single, $\sim 1\,\mu$m long, protein that extends across the half sarcomere, forming a net-like structure about the thick and thin filaments, anchoring at the Z disc (or line) (Fig. 5.7), extending as a spring across the I band, and then binding along the thick filament all the way to the M band. (The schematic showing titin in Fig. 5.18 is overly simple.) Each titin molecule is bound to the titin molecule in the adjacent sarcomere at the Z disc and to the titin in the opposite half sarcomere at the M band to form a contiguous filament along the myofibril. The elastic and viscoelastic properties of titin are due to that part in the I band.

Only the titin section in the I region is fully extensible. This section has a PEVK domain, which includes proline (P), glutamate (E), valine (V) and lysine (K) amino acid residues, flanked by immunoglobulin-like (Ig) sections (and sometimes by other sections as well), one which is proximal (to the Z disc anchor at the edge of the sarcomere) and one that is distal; these include on the order of 100 folded Ig domains linked in series.

More generally, there are N2A titin isoforms (types of titin) found in skeletal muscle sarcomeres and N2B and N2BA isoforms in cardiac muscle sarcomeres, the latter being relatively longer and more compliant within the I band. They all have PEVK sections flanked by Ig sections, but there are also N2A sections in N2A titin, N2B sections in N2B titin, and both N2A and N2B sections in N2BA titin (Fig. 5.35).

These four regions have different mechanical properties. The N2A/N2B and PEVK segments are flexible random-coil like sequences with different flexibility, and they confer an elastic response. The proximal and distal Ig domains can straighten and

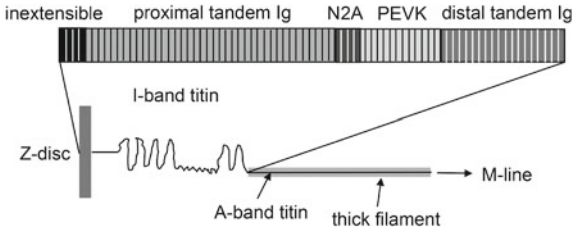


Fig. 5.35 Schematic of the I band region in titin in skeletal muscles (for the skeletal isoform N2A). (Based on [22])

can also unfold and refold, with there being more mechanical resistance to this in the distal region, and they confer a viscoelastic response, as well as an elastic response. In cardiac titin, the characteristic unfolding rate for Ig domains is $\sim 3.3 \times 10^{-3}$/s and the folding rate is ~0.3/s [45]. The folding/unfolding energies (~4 pN) and distances of a single Ig domain in titin are of the same order as those for a thick and thin filament crossbridge, as is seen by examining a single titin molecule. Within the normal physiological regime, under tension titin extends first by straightening interdomain linkers in the Ig regions and at higher forces by elongation of the N2A/N2B and PEVK segments. Only for much larger forces, usually above the physiological range, do the proximal and then the distal Ig domains extend; such unfolding may be a way to prevent muscle damage at high levels of force.

For each type of titin, the different sections each have different lengths in different specific type of muscles. This difference in the composition of the titin filament in different muscles is responsible for the varying passive stiffnesses of different muscles, with shorter forms of it (isoforms) leading to stiffer responses and higher passive tension upon extension and longer forms leading to more compliant responses and smaller passive tension. Titin may also serve as a molecular ruler of sarcomere length.

5.10 Coordination of Muscles

Many muscles are involved in most types of motion [28–30]. Sometimes they work together and are *agonists* or *protagonists* or help each other in other ways and are *synergists* (both mostly in phase) and sometimes they are *antagonists* (and out of phase). The discussion of the muscles involved in walking, running, and jumping in Chap. 3 (Figs. 3.3, 3.4, 3.5, 3.12, 3.13, and 3.31; Tables 3.1, 3.2, 3.3) and earlier in this chapter provides several examples of both protagonists (e.g., several flexors of the same joint) and antagonists (e.g., extensors/flexors of the same joint, etc.). Coordinated muscular activity during each phase of the walking cycle is shown for the three sagittal joints in Figs. 3.12 and 3.13.

Figure 5.36 shows the forces developed in the soleus and gastrocnemius muscles in the upper calf in a walking cat as a function of time during a step. They are both involved in this motion and need to work together. However, they serve different functions, the soleus (with more ST muscles) for stability and the gastrocnemius (with more FT muscles) for vertical force, and as such are not phased exactly the same. This is evident in Fig. 5.37, which plots how the soleus force evolves as a function of the gastrocnemius force, with time as an implicit variable. If they worked exactly in phase, the three sets of force plots in Fig. 5.36 would have the same shape. However, the force is developed in the gastrocnemius first during a step, and then a bit later the force in the soleus is developed. Also, as the speed of the walk increases, the force developed in the gastrocnemius increases relative to that developed in the soleus. The force plot in Fig. 5.38 shows

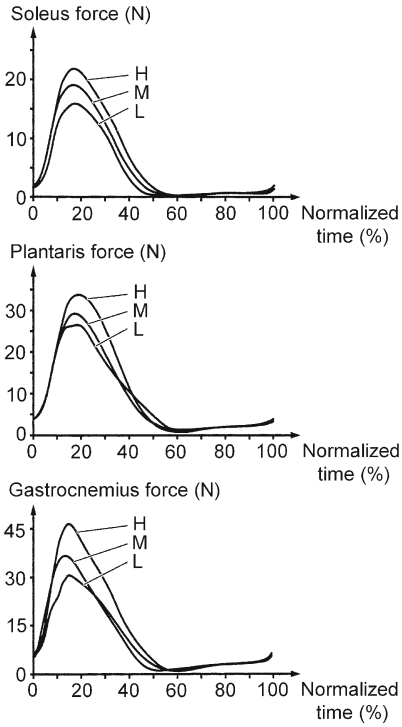


Fig. 5.36 Force versus time curves for three synergistic muscles in the ankle plantar flexor group in a cat walking at 1.2 m/s, with mean (M), highest (H), and lowest (L) forces over 43 step cycles. (From [59]. Copyright Wiley. Reproduced with permission. Also see [30])

that the tibialis anterior muscle (Fig. 3.5) and the soleus work out of phase during a step.

Bicycling is another example where the timing of muscle activation is important [16]; it is also an example where the two-joint nature of some muscles proves to be very important (see Fig. 3.2). At the beginning of the downstroke (Fig. 5.39a), the force on the pedal needs to be downward and forward, to produce extensor torques about the hip and the knee. The two-joint rectus femoris is activated then. It partially counters the effect of the activated gluteus maximus muscles, to keep the torque about the hip relatively small, and adds to the effect of the vasti to make the net extensor torque about the knee large. (This is helped by the larger moment arm of the rectus femoris about the knee than about the hip, as is seen in the figure.) At the middle of the downstroke (Fig. 5.39b), the force on the pedal needs to be downward and backward. Activation of the two-joint hamstrings contributes to the net positive extensor torque about the hip and the net flexor torque about the knee, which are needed in this stage.

Coordination within complex muscles is also needed. Problem 5.3 addresses the resultant force on the humerus from the different parts of the pectoralis major muscle activated in phase (Fig. 5.4).

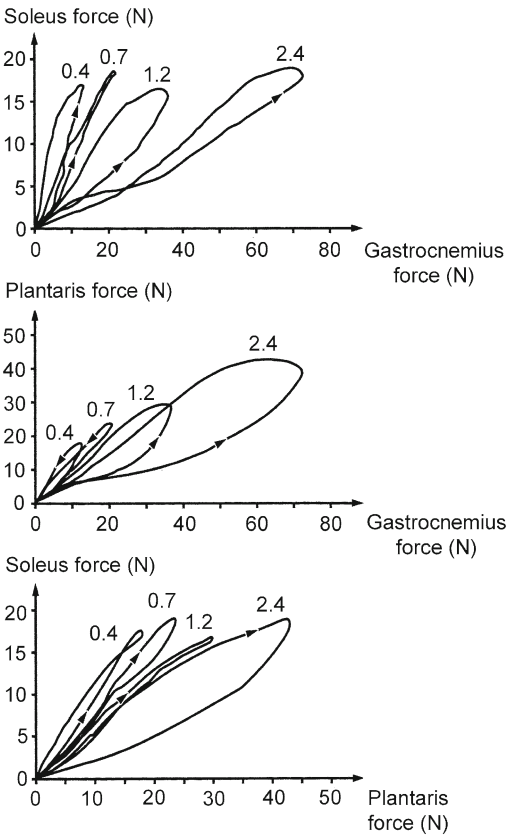


Fig. 5.37 Force versus force curves for protagonist muscles, showing force sharing for three synergistic muscles in the ankle plantar flexor group in a cat walking at 0.4, 0.7, and 1.2 m/s and trotting at 2.4 m/s, averaged over 10 cycles. (From [59]. Copyright Wiley. Reproduced with permission. Also see [30])

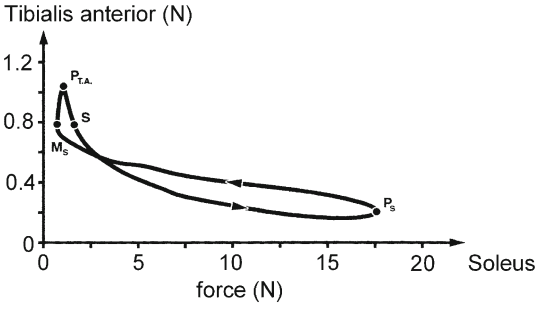


Fig. 5.38 Force versus force curves for the antagonist pair of soleus and tibialis anterior muscles in a cat walking at 0.7 m/s. (From [59]. Copyright Wiley. Reproduced with permission)

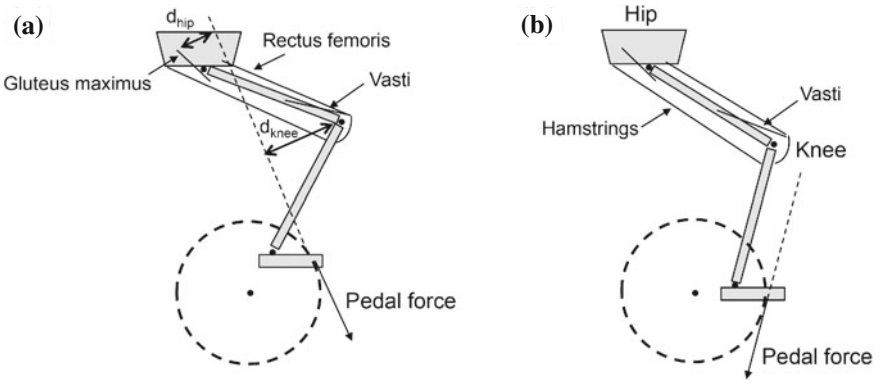


Fig. 5.39 Model of muscles crossing one or two joints that contribute at different stages of cycling. (Based on [16])

5.11 Muscle Strength and Evolution

Strength is usually measured in one of three ways: (1) the maximum force that can be exerted during an isometric contraction (the whole muscle length remains fixed, but the muscle fibers shorten), which is called the maximal voluntary contraction (MVC); (2) the maximum load that can be lifted once, the 1-repetition maximum load (1RM load); and (3) the peak torque during an isokinetic contraction (with shortening or lengthening) [16]. (In such resistance testing and training a 1RM load is defined as the highest load that can be moved, and it can be moved only once, a 10RM load is one that can be moved at most 10 times, and so on.) Fig. 5.40 shows the knee angles, forces, and torques during idealized knee exercises for these three modes. The MVC depends mostly on muscle size, while the latter two methods depend on muscle size and the coordination of the excitation of the muscles. It is not surprising that the more repetitions (r) in a set the smaller the maximum load (rRM) (Fig. 5.41), approximately as $r\text{RM} = 1\text{RM}(1.0278 - 0.0278r)$ [7].

5.11.1 Increasing Strength with Training

The PCA of a muscle increases with exercise. This increase is mostly due to increases in the cross-sectional area of individual muscle fibers, which is called *hypertrophy*. The formation of new fibers, *hyperplasia*, was once thought not to contribute to this increase at all, but now a few—but not all—studies have found evidence of it in humans. With continued training at a given load, hypertrophy continues until the muscle strength adjusts to the load.

Muscles can be strengthened by overcoming loads by muscle contraction, which is concentric exercise, as in weight lifting. The slow, controlled return movement to such concentric exercise of a given muscles utilizes eccentric contraction of the

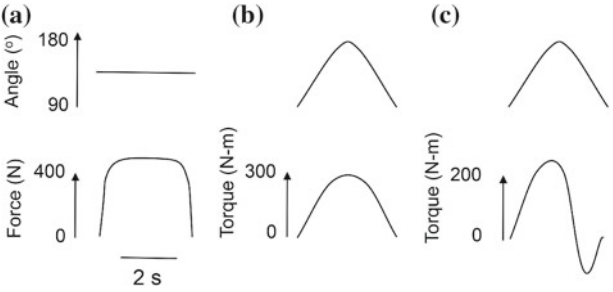


Fig. 5.40 Idealized performance of measuring strength and training knee extensor muscles for (**a**) maximal voluntary contraction (MVC), (**b**) a 1-repetition (1RM) liftand (**c**) isokinetic contraction, showing knee angle and knee force or torque. Complete knee extension is 180°. The second half of the isokinetic motion involves the knee flexor muscles. (Based on [16])

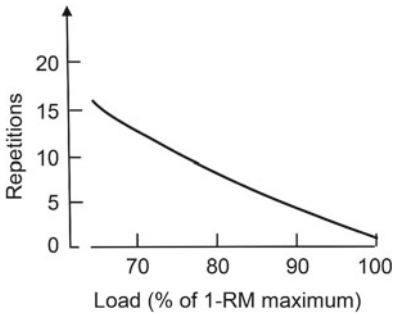


Fig. 5.41 The maximum load that can be lifted decreases with more repetitions in the set. (Based on [16, 64])

antagonist muscle; this eccentric exercise strengthens these antagonist muscles, but can lead to muscle damage. For a constant load, such as that provided by free weights, the maximum load that can be used during training (or lifting, etc.) over a large range of joint angles corresponds to the lower forces achieved by muscles much shorter and longer than the optimal length (Fig. 5.19). This does not provide very satisfactory training because the load is well below the loads that could be overcome near the optimal muscle length. Some training systems vary the load with joint angle to train the muscle optimally for the full range of joint rotation. (Can training also be improved by exercising the muscle near its optimal length with less rotation about the joint?)

In isometric exercise, the muscles try to contract against a unmoving load, so there is no actual muscle contraction; such exercise does not strengthen the muscle across its range of motion, affects only slow twitch muscle fibers, and does not provide cardiovascular or flexibility training. Training using isometric contractions does not translate well into gains of strength in activities where the muscle length changes, and is used mostly for rehabilitation. Exercise under conditions of constant muscle tension (isotonic) and with a constant rate of motion (isokinetic) are less common. Details about training can be found in [2, 8, 16, 54].

Optimizing the development of strength occurs with few repetitions at high resistance (6RM or fewer). Optimizing the development of muscular endurance occurs with many repetitions at low resistance (20RM or more). Optimizing muscle size occurs for loads between 6RM and 12RM, with short <90 s intervals between the (greater than three) repetition sets. For more details see [16, 18, 43, 54, 80].

5.11.2 Muscle Evolution with Age

In normal development, the number of muscle fibers increases in the fetus and briefly in the baby after birth, doubling in the four months after birth [2]. Slow and fast twitch muscles begin to differentiate in the fetus roughly seven months after conception, and there are roughly equal numbers of slow twitch (type I) and fast twitch (type II) fibers at birth. The diameters of muscle fibers increase during child growth (Fig. 5.42). Reaction times are slow in young children and become faster during growth, in part due to accelerated decision making (Fig. 5.43).

With advancing age comes less strength, faster fatigue, slower reactions, and poorer control of posture [2, 14, 16]. For some muscles, significant decreases in strength begin at about age 60 (Fig. 5.44a), while for others it begins at about age 20 and continues (Fig. 5.44b). In women aged 20–89, the maximum torque in knee

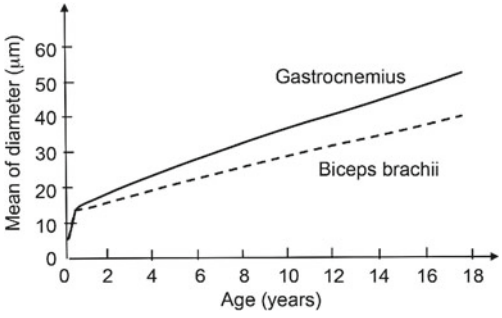


Fig. 5.42 Growth of muscle fiber diameter during childhood. (Based on [2])

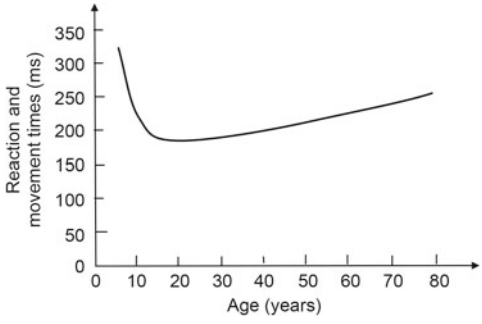


Fig. 5.43 Schematic showing how the reaction and movement times vary with age. (Based on [2])

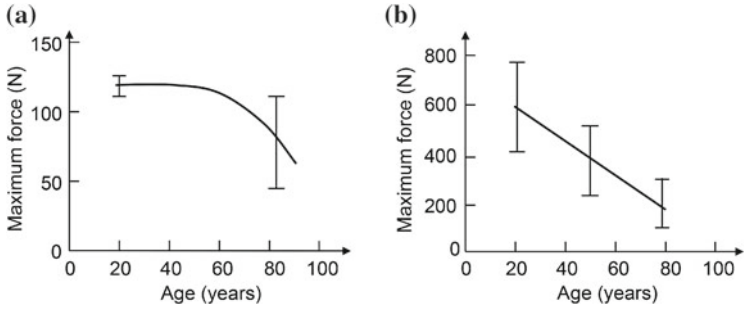


Fig. 5.44 Schematic showing how the average MVC muscle strength decreases with age for the (**a**) hand muscle adductor pollicis and (**b**) quadriceps femoris muscles, with scatter in the individual measurements also shown. (Based on [16])

extensor muscles (in N-m) decreases with age (in years) as $224 - 1.79(\text{age})$ and the reaction time (in ms) increases as $153 + 0.57(\text{age})$ [35]. The loss of strength is usually larger than the corresponding loss of muscle mass (e.g. strength loss of ~20–30% for ~15% loss in thigh muscle mass [23]), likely because of the decrease in the number of motor neurons, the reduction in the maximum rate that action potentials can be discharged (Chap. 12) during rapid contractions, and other neural effects with aging. Moreover, the loss in cross-sectional area and mass of Type II fibers is greater than for Type I fibers (e.g. ~39% vs. ~19% comparing the vastus lateralis of 25 and 67 year-old women [16]), which means that these weaker muscles also respond slower. Figure 5.43 shows that reaction and movement times lengthen as adulthood progresses.

5.11.3 Muscle Fatigue

The analyses of statics and motion in Chaps. 2 and 3 assumed muscle forces do not change with time, i.e., there is never muscle fatigue. The Rohmert curve in Fig. 5.45 characterizes the hyperbolic decrease in the endurance time of the voluntary control of a muscle with the exertion level, which is expressed as the % of maximum force level. Although not indicated in this figure, some fatigue is also expected below 15% exertion level. (Figure 5.41 shows another side of fatigue.) Metabolic changes in the muscle and impaired activation both contribute to muscle fatigue. Force can decline due to several metabolic factors that hinder the formation and detachment of crossbridges, including the formation of lactic acid, the increase in phosphate, and the decrease in the PCr (phosphocreatine) in the fibers (see Chap. 6); the depletion of ATP is not dominant. The increase in intramuscular pressure to that above the maximum blood pressure (systolic pressure, Chap. 8) restricts blood flow, which limits oxygen flow (i.e., ischemia) to the muscle and the removal of waste products. In some cases blood restriction begins at 10–20% of the maximum voluntary contraction level; some muscles are ischemic above 50% of this level. Impaired activation of muscles can

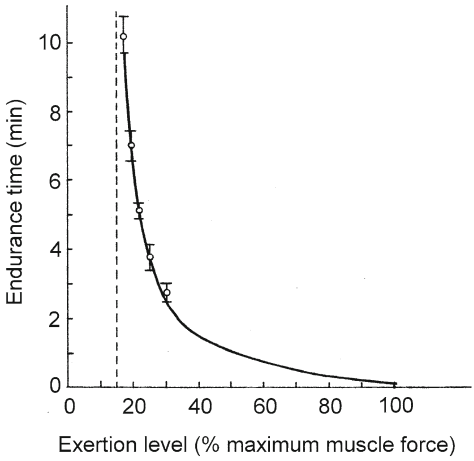


Fig. 5.45 Rohmert curve for static muscle endurance, with ±1 SD (standard devation) points shown. (From [9], adapted from [72]. Reprinted with permission of Wiley)

be due to several factors, such as central fatigue—one's motivation—and peripheral fatigue—the failure of the neural control of muscles. Type I and IIA muscle fibers are less resistant to fatigue because they are surrounded by a rich network of capillaries. Muscles used to maintain posture, such as the soleus muscles in the lower leg, contain a large fraction of type I fibers. Muscles that have a short burst of extreme activity, such as the gastrocnemius muscle, have less need to be resistant to fatigue.

5.12 Summary

The function of muscles can be understood by analyzing their substructure in terms of fibers, sarcomeres, and the filamentary structure of the sarcomeres. The forces developed by a muscle can be modeled to be a function of overall muscle length and contraction speed. Mechanical models, similar to those developed in Chap. 4, can be used to examine the consequences of the electrical stimulation of muscles. On a nanoscopic basis, forces developed by muscles can be understood and modeled in terms of the crossbridges between the myosin and actin filaments and in terms of the passive response of titin.

Problems

Muscles and Forces

5.1 There is an average normal vertical force of 1,710 N (2.7× body weight) on the one foot of an athlete in contact with the ground during running, but an average of only 715 N on each foot (and 1,430 N total vertical force with both feet = 2.3× body weight) when this same athlete does a vertical jump. Is this difference possible? Qualitatively why is it reasonable or unreasonable? (Consider only

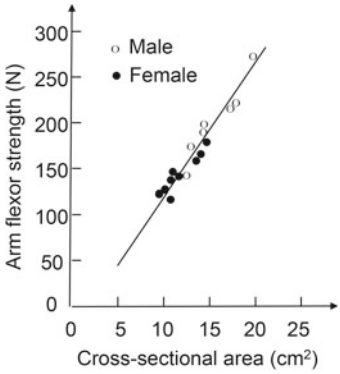


Fig. 5.46 The strength of the arm flexor muscles for males and females versus muscle cross-sectional area. (Using data from [39].) For Problem 5.2

how the ranges of muscle lengths in these activities contribute to these observations. References [3] and [4] suggest that the stimulation of the muscles during stretching that occurs immediately prior to contraction in running, but not in jumping, contributes to this observation. This is also consistent with the observation that down-and-up action prior to a high jump leads to higher jumps.)

5.2 (a) Do the data presented in Fig. 5.46 suggest that the strength of muscles varies linearly with their cross-sectional areas, as is assumed in the text?
(b) If so, what is the proportionality constant?
(c) How does it differ for males and females?

5.3 The fibers in the pectoralis major muscle converge on the proximal part of the humerus. With the humerus pointing down, the many fibers can be modeled as a clavicular component exerting a force of 224 N at an angle 0.55 rad above the horizontal and a sternal component (in the same plane and with the same timing) exerting a force of 251 N at an angle 0.35 rad below the horizontal. Find the resultant force and angle [16].

5.4 (a) Show that the physiological cross-sectional area (PCA) of a muscle for a parallel-fibered muscle is PCA $= (m/Ld)\,\text{cm}^2$, where m is the mass of the muscle fibers, d is the density of muscles ($1.056\,\text{g/cm}^3$), and L is the length of the muscle fibers.
(b) Repeat this for pinnate muscles, showing that PCA $= (m\cos\theta/Ld)\,\text{cm}^2$, where θ is the pinnation angle (the angle between the long axis of the muscle and the fiber angle).

5.5 Check to see if the two relations in Problem 5.4 are consistent with the muscle data given in Table 5.4.

5.6 Let us estimate how much weight a person can lift above his head, with extended arms.
(a) Use the data for the 70 kg person in Chap. 1, and specifically the volume of skeletal muscle. Let us assume that all of these muscles are arranged as an upright right circular cylinder of height equal to the sum of the lengths of the person's

legs, torso, and arms. Use the resulting cross-sectional area and assume that skeletal muscle can exert a force of 20 N/cm^2 to find the mass that the person can lift.
(b) Is this value reasonable? If not, describe how each of the following factors may or may not contribute to an unreasonable result: (i) His arms are narrower than his legs, so the cross-sectional area of muscle is not uniform and the limiting factor would be the weakest (i.e., narrowest) link. (ii) The muscles are not the length assumed and should be taken to be no longer than a fraction of the length of an arm or leg segment. Consequently, there are several muscles in series. (iii) Not all skeletal muscle in the body is designed to contribute to this lifting. (iv) Not all of the muscles involved are parallel fibered muscles. (v) The 20 N/cm^2 value is inaccurate.
(c) Now re-evaluate the problem for female and male world-class weightlifters for their world-record lifted masses. (Use the data given in the text.) What fraction of their skeletal muscle cross-sectional area (assumed to be uniform here) effectively contributes to the lift?

5.7 Here are some data for world weightlifting records (in 2014 for clean-and-jerk) for people of different masses: for men, 169 kg lifted (by a 56 kg man), 182 kg (62 kg), 198 kg (69 kg), 210 kg (77 kg), 218 kg (85 kg), 233 kg (94 kg), and 238 kg (105 kg); and for women, 121 kg (by a 48 kg woman), 131 kg (53 kg), 141 kg (58 kg), 143 kg (63 kg), 158 kg (69 kg), and 163 (75 kg). How does the lifted weight vary with body weight? (Do this separately for men and women.) Does it follow a power law? Is it sublinear? How can you explain this dependence?

5.8 How can a measurement of force versus muscle length be made with the muscle fixed at either end (isometric conditions), as in Fig. 5.19? Is there an inconsistency here?

5.9 Let us say that a fusiform muscle of length L and PCA is attached to bones by tendons of length $0.9L$ and cross-section area f(PCA) ($f \ll 1$) on either end. This total length of $2.8L$ is kept fixed as the muscle contracts by 20% and develops a force corresponding to 15 N/cm^2. Use the data in Table 4.2.
(a) Find the strain in the tendons. How does it compare to the UPE for tendons?
(b) Relate f to Y for the tendon.
(c) Using the value of Y from the table, compare the diameters of the muscle and tendons.
(d) Find the stress in the tendons. How does it compare to the UPE for tendons?

5.10 (advanced problem) Show that the two models in Fig. 5.14 are mathematically equivalent, with $c_1/(k_1 + k_2) = c_2/k_4$; $k_1k_2/(k_1 + k_2) = k_3$; $k_2 = k_3 + k_4$; and $(k_2/(k_1 + k_2))T_{G1} = T_{G2}$ [53].

5.11 The maximum power that can be obtained from skeletal muscles is estimated to be ~250–500 W/kg for isotonic conditions and 1.75× this with pre-stretching [25]. Assume that during baseball pitching the work done during humerus rotation is ~132 J over 0.034 s. The shoulder muscles involved in this rotation are thought to have a mass as high as of 2.16 kg in large people.
(a) What is the mechanical power associated with this shoulder rotation?

(b) Determine this power per unit mass of shoulder muscle and compare it to the maximum power derivable from pre-stretched muscle.
(c) If the power per mass available from the muscle turns out to be smaller than this measured increase in kinetic energy per unit time, explain how this could be so [63].

5.12 Repeat Problem 5.11, now including the spreads in performance as given by the standard deviations, with work done during humerus rotation being $132 \pm 52\,\mathrm{J}$ over $0.034 \pm 0.013\,\mathrm{s}$ [63].

5.13 It is known that maximum muscle forces are $1.75\times$ larger when the muscles are pre-stretched than under isotonic conditions. In the text and in Problem 5.11, it was assumed that this same factor applies when comparing the maximum specific power from muscles for these two conditions. What does this assume?

Muscle Tension Versus Length or Time

5.14 Why do you suppose that Fig. 3.54 plots the total length of the muscles and tendons?

5.15 Use the data in Fig. 3.54 to determine whether the major leg muscles used during bicycling are always near their optimal lengths (so they always exert nearly maximum forces) or do they become much longer or shorter than these optimal lengths during parts of the cycle (so they exert forces that are much less than the maxima)? Justify your answer.

5.16 In the example of throwing a ball studied in Chap. 3, we assumed that the force developed by the biceps brachii did not depend on the angle between the upper and lower arm; this angle varied from 180–0° in the first statement of the problem and from 135–45° in the second. Use what we now know about how the contractile force of a muscle varies with its length to determine the validity of that second assumption. Assume that the muscle has a length of $L_0 = 12\,\mathrm{cm}$ when resting at 90°, and that the distance from the elbow joint to the insertion point on the radius is 4 cm (see Fig. 5.47). (You should base your conclusions on calculations of the muscle length for various angles—at least for 0, 45, 90, 135, and 180°. Ignore the length of the tendons.)

5.17 In fusiform (or parallel) muscles all of the muscle contractile force is directed along the axis of the tendon and there is a relationship between changes in muscle and tendon length that depends on conditions (isometric contraction, etc.). In pinnate and bipinnate muscles these relationships are qualitatively different. Refer to Fig. 5.48, which depicts a bipinnate muscle in relatively more relaxed, initial (solid lines) and relatively more contracted, final (dashed lines) configurations.
(a) In the initial configuration, the tendon length (to the point where the muscle is inserted into the tendon, at point B) is 8.7 cm (= 2.9 cm + 5.8 cm in the figure) and each muscle fiber is 10 cm long, and is attached to the tendon at a 30° angle. Say that there are a total of N fibers attached to the tendon near this point (with an equal

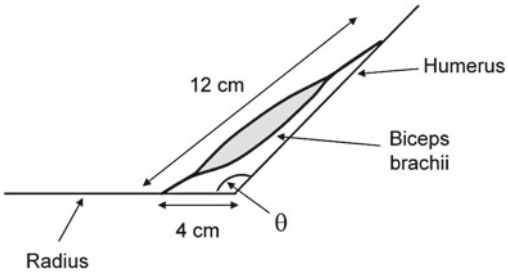


Fig. 5.47 Model of parallel fibered muscle. For Problem 5.16

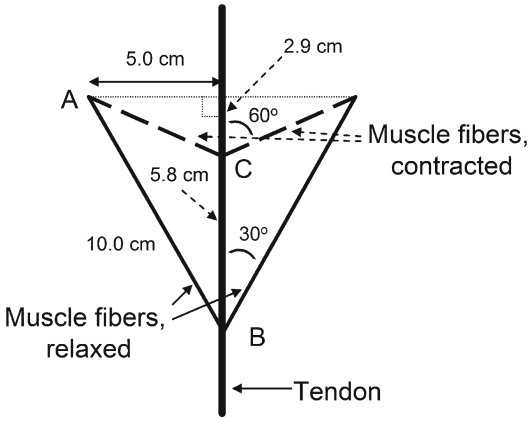


Fig. 5.48 Model of bipinnate fibered muscle. (Based on [52].) For Problem 5.17

number of fibers on both sides) and that each fiber exerts a force F_{fiber}. Find the total force transmitted to the tendon F_{tendon}. (Is there "wasted" force here?)
(b) The muscle contracts so that the tendon length (to point C) is 2.9 cm, and the muscle/tendon angle is now 60°. Find the length of the muscle (see the dashed lines). During this contraction, which has become shorter more, the muscle fibers or the tendon? Is this good or bad for muscular action? (Does this effect, along with the ability to attach more muscle fibers to this type of tendon counter the "wasted" force in part (a)?)
(c) Find the force transmitted to the tendon after the contraction in part (b) F_{tendon}, assuming that the contractile force of each muscle fiber is still F_{fiber}. How has the force transmitted to the tendon changed?
(d) Now consider the more general case, in which we acknowledge that the force exerted by a muscle fiber changes with length. Say that F_{fiber} peaks at a length $L = 8\,\text{cm}$, with $F_{\text{fiber}}(L) = F_{\max}(1 - (L - 8)^2/8)$. Sketch F_{tendon} as L contracts from 10 to 6 cm. Does it peak exactly at the length where $F_{\text{fiber}}(L)$ peaks? Why?

5.18 Let us model the decrease of muscle force F from its maximum value $F_{\max}$ at a length L_{peak} as being parabolic: $F(L) = F_{\max} - \Delta F((L - L_{\text{peak}})/L_{\text{dec}})^2$. Show that the work done by the muscle as it contracts from $L = L_{\text{peak}} + L_{\text{dec}}$ to $L = L_{\text{peak}} - L_{\text{dec}}$ is $W = W_0(1 - \Delta F/3F_{\max})$, where $W_0 = 2F_{\max}L_{\text{dec}}$ is the work that the muscle would do if the force did not change with length [55].

5.19 Sketch on one set of axes the six curves in Fig. 5.20, each as the % of maximum strength for that muscle group versus the angle of pull (which is the joint angle). The ordinate axis should range from 0 to 100% (and of course 100% represents a different maximum force for each of the curves).

5.20 (a) Show that the model we used for the torque that can develop in a joint due to muscle contraction, $\tau = \tau_{\max}((\omega_{\max} - \omega)/(\omega_{\max} + G))$, can be derived from the Hill force-velocity curve (5.21). This is expressed as a function of the rate of change of joint angle during flexion, with $\omega = -d\theta/dt$, so ω is positive when the angle is becoming smaller, and where $\omega_{\max}$ and $G \sim 3$ are constants. (We used this in the force models of the high and long jumps in Chap. 3.)
(b) Explain how the relation used for joint extension in this Chap. 3 model, $\tau = \tau_{\max}$, is a bit different than the one expected from the discussion in this chapter.

5.21 Determine the values of T and v that maximize the power output of a muscle described by the Hill force–velocity equation (5.20) and find the power output $P_{\max}$ of muscle. Take $a/T_{\max} = 0.25$; remember that $P = Tv$. (Hint: You should differentiate the expression for power with respect to v and set it equal to zero. One way to do this is to first solve for $T(v)$ from the Hill equation and express $P(v) = T(v)v$. The maximum power should turn out to be about $0.1 T_{\max} v_{\max}$.)

5.22 Use the Hill model for a muscle fiber (5.24) to show that $T(t)$ evolves as $-T - (T_{\max} + a)\ln((T_{\max} - T)/T_{\max}) = kbt$, for $\mathrm{d}L/\mathrm{d}t = 0$ (isometric condtions), when $T(0) = 0$ and $T_{\max}$ is constant [41]. Show that T eventually becomes $T_{\max}$.

5.23 Use the Hill model for a muscle fiber (5.24) to show that $T(t)$ evolves as $T_{\max} - T + (T_u + a)\ln((T_{\max} - T_u)/(T - T_u)) = kt(b + u)$, where $(T_u + a)u = b(T_{\max} - T_u)$, for a muscle fiber initially with $T(0) = T_{\max}$ that is then allowed to contract at constant speed $u = -\mathrm{d}L/\mathrm{d}t$ [41]. ($T_{\max}$ does not change.) Show that T eventually becomes T_u.

5.24 Use the Hill model for a muscle fiber (5.24) to determine the tension in a muscle fiber, initially at $T_{\max}$ that is suddenly shortened from its initial value of L_0 by L_1 at $t = 0$ [41].
(a) Show that $L(t) = L_0 - L_1(1 - \theta(t))$ and so $\mathrm{d}L/\mathrm{d}t = -L_1\delta(t)$ and (5.24) becomes $\mathrm{d}T/\mathrm{d}t = k(-L_1\delta(t) + b(T_{\max} - T)/(T + a))$. (This uses the Heaviside step and Dirac delta functions introduced in Chap. 4.)
(b) Integrate this from just before $t = 0$ to just after it (t from 0^- to 0^+), to show that $T(0^+) - T(0^-) = -kL_0$, which means that the consequence of the shortening is to decrease the initial tension from $T_{\max}$ to $T_{\max} - kL_1$, which we now call $T(0)$.
(c) Show that for $t > 0$ (5.24) becomes $\mathrm{d}T/\mathrm{d}t = kb(T_{\max} - T)/(T + a)$, and integrate it with $T(0) = T_{\max} - kL_1$ to find $T(t)$. Assume that $T_{\max}$ does not change with time. (This gives an approach to the final tension of $T_{\max}$ that is faster than observed by experiments, even with optimal curve fitting, and this illustrates one of the limitations of the Hill force-velocity curve.)

5.25 In the example of throwing a ball studied in Chap. 3, we assumed that the force developed by the biceps brachii did not depend on how fast the elbow angle θ varies with time. Figure 5.28 shows that this is not true, because the muscle length and contraction speed can both be important and can limit the tension developed in the muscle. One way to address this is by re-analyzing throwing, limiting muscle tension as indicated in this figure. Explain how this can be done numerically.

5.26 In the analysis of throwing a ball studied in Chap. 3, we assumed that the force developed by the biceps brachii may depend on muscle length but not on how fast the elbow angle θ varied with time ($d\theta/dt$). We analyze the limitations of that model by adopting an approach that is quite different from that used in Problem 5.25 (and we will not use Fig. 5.28 explicitly). Say $T_{max} = (20\,N/cm^2)PCA$, where PCA is the physiological cross-sectional area of a muscle, and that $v_{max} = 6L_0/s$ for muscles predominantly comprised of ST muscle fibers and $v_{max} = 16L_0/s$ for muscles predominantly comprised of FT muscle fibers, where L_0 is the optimal fiber length of the muscle of interest. (Because muscles are often composed of combinations of FT and ST muscle fiber, the appropriate v_{max} is often in between these limiting values.) Also assume that $a/T_{max} = b/v_{max} = 0.25$ and use the results from Problem 5.21. For each part that follows, consider the case studied in Chap. 3 of 2 in diameter biceps brachii muscles, for which the calculated throwing speed is 17.8 mph (with no gravity and $\langle \sin\theta \rangle = 1$), and take $L_0 = 10\,cm$.

(a) Find a and b, first assuming FT and then ST muscles.

(b) If the ball leaves the hand at a speed of x mph, geometry says that the speed of muscle contraction is smaller by the proportion of the distance of muscle insertion from the elbow pivot (4 cm) to the distance of the ball from the pivot (36 cm). How does this speed of muscle contraction, based on the throwing calculation, compare to the maximum muscle contraction speed (v_{max}) for FT and ST muscles? Also, how does it compare to the muscle contraction speed, for both types of muscles, at which the power generated by the muscle is maximized? (Ignore the response of the tendons. Can these be important?)

(c) The average and peak powers needed to be generated by the muscles to achieve these throwing speeds are 178 W and 356 W, respectively, for the 2 in diameter biceps using the kinematics calculations performed in Chap. 3. For both FT and ST muscles, calculate the maximum power that the muscle can generate and compare your answers with the values calculated using kinematics. Repeat this if you (incorrectly) assume that the force generated by the muscle is T_{max} independent of muscle contraction speed, so that the maximum muscle power would be $T_{max}v_{max}$ (which is clearly incorrect).

(d) Do the results in (b) and (c) cast doubt on the calculation in Chap. 3 (and why)? If so, does this totally invalidate the calculation or does it mean that after a certain muscle contraction speed is achieved the decrease in the muscle force must be included to improve the model? (Also, note that since the biceps brachii have much ST muscle fiber, the "FT" limit is not very realistic.)

Microscopic and Nanoscopic Processes in Muscles

5.27 (a) Estimate the time it takes Ca^{2+} ions to diffuse throughout a typical skeletal muscle cell if the cell has a diameter of 200 μm and the diffusion coefficient of the ions is $10^{-5}\,cm^2/s$. Assume that the release of ions occurs just outside of the cell, due to a nerve pulse arriving there, and diffusion occurs in one transverse dimension of the cell.
(b) How does this time compare to the typical reaction times of muscles?
(c) The nerve signal actually activates the sarcoplasmic reticulum that runs transverse to the outer membrane of the cell, in the Z-line borders of each sarcomere, and the Ca^{2+} ions are released there; consequently, the maximum distance the ions need to travel to diffuse across the whole sarcomere is 2 μm. How long does this take and does this time seem more reasonable?

5.28 Estimate the number of sarcomeres in your biceps brachii:
(a) along its length,
(b) across its cross-section, and
(c) in total.

5.29 When lifting a weight, a muscle goes from being 25% shorter than its resting length to 25% longer than it. If the resting length of the sarcomere is 2.5 μm, how many 11-nm crossbridge power strokes occur in each sarcomere during the lift?

5.30 Figure 5.33 is the force versus displacement curve of a single myosin molecule interacting with an actin molecule during a powerstroke, as measured by "optical tweezers." Since work is the integral of force over distance, estimate the work done by the myosin molecule during one powerstroke. (Use the dotted line and find the area in the triangle.) If the energy available from ATP hydrolysis in a muscle cell is 10^{-19} J, calculate the efficiency of a myosin powerstroke in using the energy from ATP hydrolysis. This estimate uses the mean force. The efficiency is ~1.75× larger if a less conservative estimate is made using the highest forces measured [17].

5.31 (a) If in a single actin–myosin crossbridge 4 pN is generated per crossbridge and the power stroke distance is 11 nm, as in Fig. 5.33, show that the mechanical work done is 22 pN–nm. Express this in J and eV (1 eV (electron volt) = 1.6×10^{-19} J) per crossbridge, and in kcal/mole (for a mole of crossbridges).
(b) If the hydrolysis of ATP releases 14 kcal/mole (as is seen in Chap. 6) and the hydrolysis of one ATP molecule activates one crossbridge, what fraction of the available energy is used in mechanical work?

5.32 (advanced problem) In the Huxley sliding filament model, assume $f(x) = 0$ for $x < 0$, $f_1 x/h$ for $0 \le x \le h$, and 0 for $x > h$, and $g(x) = g_2$ for $x \le 0$ and $g_1 x/h$ for $x > 0$. Use (5.34) to show that $T_{max} = Nkh^2 F_1/2$, where $F_1 = f_1/(f_1 + g_1)$ [41].

5.33 (advanced problem) Use the Huxley sliding filament model (5.33) in steady state with constant contraction speed v, and $f(x)$ and $g(x)$ and the other nota-

tion from Problem 5.32, to show that for these rate functions $n(x) = F_1(1 - \exp(-V/v))\exp[(2x/h)G_2(V/v)]$ for $x < 0$, $= F_1[1 - \exp([(x^2/h^2) - 1](V/v))]$ for $0 < x < h$, and $= 0$ for $x > h$, where the constant $V = (f_1 + g_1)h/2$, which has units of speed, and $G_2 = g_2/(f_1 + g_1)$ [41]. (Hint: Do this by separately solving the Huxley equation in the three spatial regions, using the appropriate forms of the rate functions, to arrive at separate solutions and then choose the undetermined constants to make $n(x)$ continuous at $x = 0$ and $x = h$. Keep only solutions that do not go to infinity in the region (i.e. are bounded) within each range (and this means that the solution for $x > h$ is $n = 0$). For the middle region use a trial function $n(x) = A + B\exp(Cx^2)$.)

5.34 (advanced problem) Use $n(x)$ and other notation from the solution to the Huxley sliding filament model for the chosen rate functions in Problem 5.33 to show that (5.32) becomes $T(v) = T_{\max}[1 - (V/v)[1 - \exp(-(V/v))](1 + (v/V)/2G_2^2)]$ [41]. This is the force-velocity equation that can reproduce the Hill force-velocity curve.

5.35 (advanced problem)
(a) In the Huxley sliding filament model prediction for $T(v)$ from Problem 5.34, T ranges from $T_{\max}$ for $v = 0$ to 0 for a given $v = v_{\max}$. Show that if $F_1 = 0.812$ and $G_2 = 3.919$, then $v_{\max} \simeq 4V$.
(b) Since $v_{\max} = bT_{\max}/a$ in the Hill force-velocity curve, show that with $T_{\max} = 4a$ (which is a common fit), then $V \simeq b$ in this application of the Huxley model [41]. This shows how the Huxley sliding filament model reproduces the limits of the Hill curve, and it is easy to show that it gives the same curve shape in between these two limits.

5.36 (advanced problem)
(a) For $f_1 = 65$/s, $g_1 = 15$/s, and $h = 10$ nm, show that the Huxley model results in Problem 5.35 predict a maximum shortening speed of 1600 nm/s for each filament pair, which means for each half sarcomere, or 3200 nm/s for each sarcomere [41].
(b) If the sarcomeres are 2.5 μm long and the muscle length is 1 cm, find how fast the muscle can contract.

5.37 (advanced problem) Plot the Huxley model results for $T(v)$ from Problem 5.35 on the same axes as the Hill force-velocity curve and compare them.

Muscle Energy, Power, Synergy, Training, and Endurance

5.38 By using reasonable estimates of muscle length, speed and force per unit area, determine whether the estimate that ~250 W are generated per kg of active muscle is reasonable.

5.39 A 75 kg person climbs stairs to a height of 50 m in 53 s. Determine the person's metabolic rate during climbing, in W and kcal/h, assuming the process in 25% efficient. How does this relate to the estimates of energy used in such motion in the text? (See Chap. 6 for more on metabolism.)

5.40 In a weight room, you exercise your biceps brachii with a constant weight by trying to lift it (i.e., rotating your lower arm about your elbow). Sketch the maximum tension versus length for your biceps brachii if
(a) The weight is so heavy that you can barely hold it as you try to lift it.
(b) The weight is so light that you can easily lift it.
For both cases, where is the weight on your sketch? Describe howis also an exampleis also an example much you are changing the length of your muscle during the "lift."

5.41 During a forearm curl exercise, your upper arm is fixed, say at 45° to the vertical and your elbow is fixed, as you rotate your forearm (and hand) about your elbow (decreasing and later increasing the elbow angle), while holding a weight in your hand.
(a) Draw diagrams showing the directions and magnitudes of the forces due to the weight and your lower arm, both near the beginning of the first phase of the curl (elbow flexion, beginning with elbow angle 180°) and the second phase of the curl (elbow extension, beginning with 25°), first ignoring the muscles that flex the elbow [16].
(b) Labeling forces and distances as needed, find the torque your arm muscles need to supply for both cases, first assuming only equilibrium.
(c) Calculate these two torques for a 160 lb, 6 ft standard human and a dumbbell weight of 30 lb.
(d) Would you expect the torques you would need to apply during a curl (with actual motion) to be greater than, the same as, or less than these values?

5.42 (a) Draw a force diagram for the forearm curl exercise as in Problem 5.41, with your hand still holding a weight such as a barbell or dumbbell (constant-load training), and determine and sketch the torque about the elbow your muscles need to supply to maintain equilibrium as a function of elbow angle. In constant-load training the magnitude and direction of the force due to the free weight do not change during the exercise cycle.
(b) Repeat (a) for a variable load exercise machine in which a force is applied (at the hand) normal to the forearm that can vary with angle (variable-load training) [16, 54]. What should that variable load be as a function of angle to maintain equilibrium as a function of elbow angle (and at static equilbrium)? How does your answer differ from that in (a)?
(c) The maximum amount of torque your muscles can supply depends on their length and consequently on the joint angle. In training you often want to push the limits of your muscles to some extent. Are the features of variable-load training preferable to those of constant-load training for weight training? Why?

5.43 Fig. 5.37 shows that as a cat walks faster his gastrocnemius develop more and more force, but his soleus muscles do not. Explain why the relative contribution of the gastrocnemius muscle increases with speed.

5.44 You repetitively extend your arm and then flex it (say as to touch your shoulder), with a period of 2 s. Sketch the forces in your biceps brachii and triceps brachii each versus time, and both versus each other (with time as an implicit parameter).

5.45 Derive a hyperbolic expression that describes the Rohmert curve (Fig. 5.45) relating between the endurance time and exertion level of a muscle.

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